

Appendix

This appendix is the formal specification of the CTCC cost–benefit methodology that backs the Methods and case studies in the main text. Each section below lists variables, units, equations, and conditional logic for one cost or benefit component so the mapping from engineering assumptions to dollar totals is transparent. All cost and benefit symbols (C , B) denote present values unless explicitly marked ,nominal.

The appendix is organized into five top-level sections. Utilities defines shared building blocks (weighted miles, discount rates, AFUDC and rate base, system constraints). Costs groups components into four buckets (hard, soft, risk, emissions). Benefits groups components into remedial and enabling buckets plus displacement reporting. Reporting Framework assembles bucket totals, net benefit, incrementals, and BCR perspectives. Revenue defines capital recovery and the full revenue requirement (utility–ratepayer transfer).

A Utilities

This section defines shared mathematical building blocks that multiple cost and benefit components depend on. Each utility is defined once and referenced throughout the cost and benefit subsections in subsequent sections: weighted miles (terrain-adjusted distance), financial parameters (discount rates), AFUDC and rate base, and system constraints and capacity relief.

A.1 Weighted Miles

This subsection defines weighted miles, the difficulty-adjusted line miles used for build costs and other components that use terrain-adjusted distance (used by build costs §B.1.1, ROW capital §B.1.2, and environmental mitigation §B.1.3). The terrain set is the set of terrain types: Forested, Scrubbed Flat, Wetland, Farmland, Desert/Barren, Urban, Rolling Hills, Mountain, Subsea. Only terrains with $M_{\text{terrain}} > 0$ are used in sums below and elsewhere in this appendix. Default terrain multipliers λ_{terrain} are sourced from the WECC Transmission Expansion Planning cost study (Rawlins et al. 2014), which derives construction difficulty factors from utility-reported project data (Black & Veatch, 2014). The wetland multiplier (2.0) is updated from MISO MTEP25 matting and construction-difficulty adders (MISO 2025b), superseding the WECC value (1.20) which relied on a single California data point.

Variables

Variable	Description
M_{terrain}	Miles of a specific terrain type (miles). Terrain set as defined in this subsection. Only terrains with $M_{\text{terrain}} > 0$ are used in sums.
λ_{terrain}	Cost multiplier by terrain type (dimensionless, ≥ 1). Same terrain list as M_{terrain} .
M_{total}	Total (raw) miles (miles). Sum of M_{terrain} over all terrains.
$M_{\text{weighted,terrain}}$	Weighted miles for that terrain type (miles). $M_{\text{weighted,terrain}} = M_{\text{terrain}} \times \lambda_{\text{terrain}}$ for that terrain.
$M_{\text{weighted,total}}$	Total weighted miles across all terrain types (miles). Used in Build costs (§B.1.1) for conductor/structure/converter variable costs.
λ_{average}	Average terrain multiplier for the project (dimensionless). $\lambda_{\text{average}} = M_{\text{weighted,total}}/M_{\text{total}}$ when $M_{\text{total}} > 0$.

Equations

1. Weighted miles for each terrain type

$$M_{\text{weighted,terrain}} = M_{\text{terrain}} \times \lambda_{\text{terrain}} \quad \text{for that terrain type.}$$

2. Total weighted miles

$$M_{\text{weighted,total}} = \sum_{\text{terrain}} M_{\text{weighted,terrain}}.$$

3. Total (raw) miles

$$M_{\text{total}} = \sum_{\text{terrain}} M_{\text{terrain}}.$$

4. Average terrain multiplier

$$\lambda_{\text{average}} = \frac{M_{\text{weighted,total}}}{M_{\text{total}}} \quad (M_{\text{total}} > 0).$$

Table 1: Default terrain multipliers λ_{terrain} .

Terrain	λ	Source
Forested	2.25	WECC 2014
Scrubbed flat	1.00	WECC 2014 (baseline)
Wetland	2.00	MISO MTEP25
Farmland	1.00	WECC 2014 (baseline)
Desert/barren	1.05	WECC 2014
Urban	1.59	WECC 2014
Rolling hills	1.40	WECC 2014
Mountain	1.75	WECC 2014
Subsea	1.00	Marine costs separate

A.2 Financial Parameters

This subsection defines discount rates, inflation, and base year used across CTCC. The real WACC ($r_{\text{WACC,real}}$) discounts all market-valued costs and benefits; the social discount rate (r_{social}) discounts emissions externalities (§B.4). The default social discount rate of 2% follows the Ramsey discounting framework in Office of Management and Budget (2023) and Rennert et al. (2022).

Variables

Variable	Description
π	Annual inflation rate (decimal). Used in Fisher link for real WACC. Default 3%.
Y_{base}	Base year (year). Reference year for all present values and cost parameters. Default 2025; all costs expressed in Y_{base} constant dollars.
$r_{\text{WACC,nom}}$	Nominal weighted average cost of capital (decimal). Direct input. Converted to $r_{\text{WACC,real}}$ via Fisher link; not used directly in any calculation. Default 7.5%.
$r_{\text{WACC,real}}$	Real weighted average cost of capital (decimal). Used for societal PV of costs and benefits. Calculated via Fisher link.
r_{social}	Social discount rate (decimal). Used for externality costs (emissions, expected wildfire cost, expected outage cost).
$g_{\text{construction}}$	Construction cost escalation rate (decimal). Default 2.5%/yr nominal. Applied during delay (§B.1.1). Source: MISO MTEP25 (MISO 2025b).
g_{benefit}	Benefit price escalation rate (decimal). Real annual escalation applied to congestion, curtailment, and delivered-energy benefit prices. Default 1%/yr real. At 0%, reduces to the flat-price case. Source: Chang, Pfeifenberger, and Hagerty (2013) (1% real tail-extension practice); California Independent System Operator (2025) (0% real current practice); sensitivity 0–4%.
g_{rent}	ROW rent escalation rate (decimal). Real annual escalation applied to operational ROW rent (§B.2.3). Default 0%/yr real. BLM adjusts at the IPD-GDP rate ($\approx$ inflation); no above-inflation trend (Bureau of Land Management 2026a; U.S. Government 2024).

Equations

1. Real WACC (Fisher link)

$$r_{\text{WACC,real}} = \frac{1 + r_{\text{WACC,nom}}}{1 + \pi} - 1.$$

When each rate is used

- $r_{\text{WACC,nom}}$: Specified as a model input; not used directly in any calculation. All calculations—including AFUDC—use $r_{\text{WACC,real}}$ derived via the Fisher link above.
- $r_{\text{WACC,real}}$: Societal PV of capital, O&M, insurance, ROW rent, energy loss costs, benefits, revenue; delay and congestion/curtailment delay costs.
- r_{social} : Societal PV of externality costs: emissions (§B.4), expected cost of wildfires (§B.3.1), expected cost of outages (§B.3.2).

- g_{benefit} : Real annual growth in benefit prices (congestion, curtailment, delivered energy) and welfare-consistent energy valuation (thermal loss cost, delay costs). Applied from COD over T_{lifetime} in §C.1.1, §C.1.2, §C.2.1, §B.2.5; applied over $T_{\text{delay}} + T_{\text{construction}}$ in §B.2.7, §B.2.8.
- g_{rent} : Real annual growth in ROW rent. Applied from $t_{\text{rent,start}}$ over T_{rent} in §B.2.3.

Cost normalization. All cost parameters in this appendix are expressed in Y_{base} constant dollars (default 2025). Source values from earlier or later years are escalated or deflated using the BLS Consumer Price Index for All Urban Consumers (CPI-U, Series CUUR0000SA0, annual averages, 1982–84 = 100). Table 2 lists the adjustment factors applied. Items accepted on author judgment (wildfire severity composite, urban vegetation management) are flagged in their respective subsections.

Table 2: Base-year normalization factors (CPI-U annual averages).

Source year	CPI-U	Factor → 2025\$	Parameters
2007	207.342	1.5525	Habitat conservation credits
2008	215.303	1.4951	Vegetation management (desert, farmland)
2020	258.811	1.2438	Social cost of CO ₂ , SO _x , NO _x
2021	270.970	1.1880	Outage redispatch cost
2022	292.655	1.0999	Vegetation management (rolling hills)
2023	304.702	1.0564	Overhead conductor, non-overhead build costs, converter stations
~2023.5	309.194	1.0411	Wildfire severity (composite)
2024	313.685	1.0262	Value of lost load tiers, vegetation management (forested, wetland, scrubbed)
2025	321.900	1.0000	Target base year
2026	330.900	0.9728	BLM ROW rents and hold costs (deflation)

Overhead structure costs are sourced directly from MISO MTEP25 (MISO 2025b) at 2025\$ and do not require index adjustment. BLM acquisition costs (15 zones) are exempt from normalization: they are administrative zone values set by BLM regulation, unchanged since 2016, and represent actual schedule amounts payable (Bureau of Land Management 2026b). Environmental mitigation base costs for forested (\$5,486/acre) and mountainous (\$7,132/acre) terrain use MISO MTEP25 clearing costs directly at 2025\$ (MISO 2025b). Remaining terrain base mitigation costs (scrubbed flat, wetland, farmland, desert/barren, urban, rolling hills: \$800–\$5,000/acre overhead; underground values scaled 1–2× for trench/restoration scope) are order-of-magnitude estimates anchored to the MISO Cost Estimation Guide’s light-vegetation terrain/grading floor of \$265/acre and commercial ROW clearing benchmarks of \$700–\$2,500/acre. No published terrain×construction-type matrix exists for transmission environmental restoration; per-terrain differentials reflect author judgment informed by relative clearing difficulty. Underground wetland (\$10,000/acre) reflects dewatering and wetland-restoration requirements beyond surface clearing. Habitat conservation credits default to \$49,188/acre (2025\$), sourced from the Ecosystem Marketplace national average (Madsen, Carroll, and Moore Brands 2010). The uniform rate across terrains reflects that credit prices vary primarily by geography rather than ecosystem type (U.S. Government Accountability Office 2024).

The construction cost escalation rate $g_{\text{construction}}$ is applied during the delay period (§B.1.1) to reflect that deferred projects face higher material, labor, and equipment costs when they eventually break ground.

A.3 AFUDC and Rate Base

This subsection defines allowance for funds used during construction (AFUDC) and rate base. AFUDC compounds base-year spending on construction work in progress (CWIP) to the commercial operation date (COD). Rate base at COD is the sum of capitalized AFUDC-eligible capital: build, ROW capital (acquisition), and environmental mitigation. The real WACC from §A.2 is used as the AFUDC rate, because all CTCC costs are expressed in base-year (real) dollars. This is equivalent to the traditional regulatory approach of compounding nominal-dollar costs at the nominal WACC and deflating to base-year dollars. Costs not AFUDC-eligible (e.g. delay, O&M, ROW rent, wildfire/outage) do not enter rate base.

Variables

Variable	Description
i	Component index (index). $i \in \{\text{build, ROW, env mitigation}\}$.
C_i^{nom}	Spending (pre-AFUDC) for component i , base-year (\$). In base-year real dollars.
T_{delay}	Delay years (years). From project technical details.
$T_{\text{construction}}$	Construction years (years). From project technical details.
w_y	Spending weight for construction year y (dimensionless). From S-curve profile; $\sum_{y=1}^{T_{\text{construction}}} w_y = 1$.
r_{AFUDC}	AFUDC rate (annual) (decimal). From §A.2: $r_{\text{AFUDC}} = r_{\text{WACC,real}}$.
AFUDC_i	AFUDC allowance for component i (\$). $C_i^{\text{cap}} - C_i^{\text{nom}}$.
C_i^{cap}	Capitalized cost at COD for component i (\$). $C_i^{\text{cap}} = C_i^{\text{nom}} + \text{AFUDC}_i$.
RB_{real}	Rate base at COD, base-year (\$). Sum of C_i^{cap} over build, ROW, env mitigation. Already in base-year dollars (AFUDC compounds at real rate).

Equations

1. AFUDC rate.

$$r_{\text{AFUDC}} = r_{\text{WACC,real}} \quad (\text{from §A.2}).$$

2. Capitalization to COD. All AFUDC-eligible capital costs are incurred during the construction period. Each year’s fractional spend is compounded from its midpoint to COD using S-curve weights. See paragraph 2b.

Delay-period spending. ROW holding costs (annual option fees to maintain land access during the permitting period) are incurred during the delay period but are not AFUDC-eligible: they are operating expenses during delay and do not enter rate base. Only ROW acquisition is AFUDC-eligible capital. The cost timing pattern configuration governs the delay/construction split for AFUDC-eligible costs; the default is 100% during construction.

2b. Timing methodology (year-by-year compounding). Construction spending follows an S-curve disbursement profile derived from MISO’s Transmission Cost Estimation Guide (MISO 2025a). Table 3 gives the reference profiles for 5–8 year construction durations; for other durations, the profiles are pooled into a normalized cumulative curve and resampled via piecewise linear interpolation. Each year’s fractional spend is compounded from its midpoint to COD:

$$C_i^{\text{cap}} = C_i^{\text{nom}} \sum_{y=1}^{T_{\text{construction}}} w_y (1 + r)^{T_{\text{construction}} - y + 0.5},$$

where w_y is the spending weight for year y ($\sum w_y = 1$) and $r = r_{\text{AFUDC}}$. Hence $\text{AFUDC}_i = C_i^{\text{cap}} - C_i^{\text{nom}}$.

Table 3: MISO reference construction spending profiles (Table 5-1, MISO 2025a).

Duration	Yr 1	Yr 2	Yr 3	Yr 4	Yr 5	Yr 6	Yr 7	Yr 8
5-year	1%	4%	35%	35%	25%			
6-year	1%	4%	10%	30%	35%	20%		
7-year	1%	1%	3%	15%	30%	30%	20%	
8-year	1%	1%	3%	5%	15%	25%	30%	20%

3. Rate base at COD.

$$RB_{\text{real}} = C_{\text{build}}^{\text{cap}} + C_{\text{ROW}}^{\text{cap}} + C_{\text{env mit}}^{\text{cap}}.$$

Which costs enter rate base. Build costs (§B.1.1), capital ROW (§B.1.2), and environmental mitigation (§B.1.3) are the three components in rate base. Capital recovery (§E) uses RB_{real} to compute the depreciation-plus-return component of the utility’s revenue requirement.

A.4 System Constraints and Relief

This subsection defines the system constraint a project addresses and the capacity relief it provides. Effective capacity relief ($\Delta C_{\text{effective}}$) and the Approach B congestion/curtailment split feed congestion benefit (§C.1.2), curtailment benefit (§C.1.1), delivered energy (§C.2.1), and facilitated emissions (§B.4.2).

Variables

Variable	Description
$\Delta C_{\text{effective}}$	Effective capacity relief (MW).
ϕ	Flow factor (dimensionless [0,1]). Greenfield: fraction of nameplate that relieves constraint.
C_{new}	New line capacity (MW). Greenfield and reconductoring.
C_{old}	Old line capacity (MW). Reconductoring only.
H	Constrained hours (h/year). Hours/year the transmission constraint binds.
X	Average exceedance (MW). Average MW exceedance during constrained hours.
f	Congestion fraction (dimensionless [0,1]). Fraction of constrained hours resolved via redispatch (congestion); the remainder $(1 - f)$ is resolved via renewable curtailment.
$\gamma_{\text{congestion}}$	Value of congestion (redispatch cost) (\$/MWh).
$\gamma_{\text{curtailment}}$	Value of curtailment (foregone energy value) (\$/MWh).
$B_{\text{congestion,annual}}$	Annual congestion benefit (\$/year).
$B_{\text{curtailment,annual}}$	Annual curtailment benefit (\$/year).
$B_{\text{remedial,annual}}$	Annual remedial benefit (\$/year). Sum of congestion and curtailment benefit.

Equations

1. Effective relief.

$$\Delta C_{\text{effective}} = \begin{cases} \phi \times C_{\text{new}} & \text{if greenfield} \\ C_{\text{new}} - C_{\text{old}} & \text{otherwise} \end{cases}$$

2. Approach B: single-constraint, two-price decomposition. Congestion and curtailment arise from a single transmission constraint. During constrained hours H , a fraction f of binding events result in redispatch (congestion); the remainder $(1 - f)$ result in renewable curtailment:

$$\begin{aligned} B_{\text{congestion,annual}} &= H \times f \times \min(\Delta C_{\text{effective}}, X) \times \gamma_{\text{congestion}}, \\ B_{\text{curtailment,annual}} &= H \times (1 - f) \times \min(\Delta C_{\text{effective}}, X) \times \gamma_{\text{curtailment}}, \\ B_{\text{remedial,annual}} &= B_{\text{congestion,annual}} + B_{\text{curtailment,annual}}. \end{aligned}$$

Guidance for f . The congestion fraction reflects the corridor’s dominant constraint type:

Corridor type	Typical f	Rationale
Wind corridor (CREZ-type)	0.05–0.10	Near-pure curtailment
Wind HVDC (SunZia-type)	0.20–0.40	Mostly curtailment, some load-side congestion
Offshore wind	0.10–0.30	Mostly curtailment
Load pocket (LRGV-type)	0.80–1.00	Congestion-dominated

B Costs

This section defines the four cost buckets: hard costs (C_{hard}), soft costs (C_{soft}), risk costs (C_{risk}), and emissions costs ($C_{\text{emissions}}$). The grand total cost for scenario P is $C^P = C_{\text{hard}}^P + C_{\text{soft}}^P + C_{\text{risk}}^P + C_{\text{emissions}}^P$.

B.1 Hard Costs (C_{hard})

Hard costs are one-time capital expenditures that are AFUDC-eligible and enter rate base at COD: build costs, capital ROW (acquisition), and environmental mitigation (see §A.3). These three components produce the hard cost bucket C_{hard} (§B.1.3).

B.1.1 Build costs

This subsection defines build cost: terrain-adjusted, contingency-applied, and soft-cost-adjusted conductor, structure, and converter costs. Default contingency is 20%, consistent with MISO’s planning-level estimate classification (MISO 2024b; MISO 2025b). A soft cost multiplier (default 10%) is applied after contingency to capture project management, engineering/design, and administrative costs. Build cost is AFUDC-eligible and enters rate base at COD. Component costs use weighted miles (§A.1); reconductoring excludes structure and converter.

Default calibration

Overhead conductor costs (Standard Aluminum/ACSR) derive from MISO MTEP24 Cost Estimation Guide Tables 2.2-11/12 (MISO 2024b), escalated from 2023\$ to 2025\$ via CPI-U ($\times 1.0564$). Advanced Aluminum (ACCC) conductor costs are from the same vintage, cross-checked against CTC Global proprietary data. Overhead structure costs use MISO MTEP25 steel-tower tables (2.2-2 AC, 2.2-8 DC) multiplied by structures-per-mile from Tables 3.1-2/3.1-4 (MISO 2025b), directly in 2025\$. Converter station costs derive from MISO MTEP24 Tables 2.4-1/2.4-2, escalated from 2023\$ to 2025\$. Non-overhead (underground and subsea) conductor and fixed costs derive from the IET/Mott MacDonald Electricity Transmission Costing Study (2012 GBP, converted at GBP/USD = 1.5852, escalated to 2025\$ via CPI-U).

Variable	Description
$C_{\text{conductor}}$	Conductor cost (\$). Not terrain adjusted, no contingency applied.
$C_{\text{structure}}$	Structure cost (\$). Not terrain adjusted, no contingency applied.
$C_{\text{converter}}$	Converter cost (if DC) (\$). Not terrain adjusted, no contingency applied.
$C_{\text{adj,conductor}}$	Conductor cost (\$). Terrain adjusted, no contingency applied.
$C_{\text{adj,structure}}$	Structure cost (\$). Terrain adjusted, no contingency applied.
$C_{\text{adj,converter}}$	Converter cost (if DC) (\$). Terrain adjusted, no contingency applied.
$C_{\text{adj,cont,conductor}}$	Conductor cost (\$). Terrain adjusted, contingency applied.
$C_{\text{adj,cont,structure}}$	Structure cost (\$). Terrain adjusted, contingency applied.
$C_{\text{adj,cont,converter}}$	Converter cost (if DC) (\$). Terrain adjusted, contingency applied.
$F_{\text{conductor}}$	Fixed conductor costs (\$). Conductors have both fixed and variable costs.
$F_{\text{structure}}$	Fixed structure costs (\$). Currently, structures have no fixed costs in the CTCC.
$F_{\text{converter}}$	Fixed converter costs (\$). Converters only have fixed costs.
$V_{\text{conductor}}$	Variable conductor costs (\$/mile). Miles used are terrain adjusted (§A.1).
$V_{\text{structure}}$	Variable structure costs (\$/mile). Miles used are terrain adjusted.
$V_{\text{converter}}$	Variable converter costs (\$/mile). Converters currently have no variable costs — only fixed.
$M_{\text{weighted,total}}$	Total weighted miles (miles). From §A.1.
$r_{\text{contingency,conductor}}$	Contingency multiplier conductors (dimensionless [0,1]).
$r_{\text{contingency,structure}}$	Contingency multiplier structures (dimensionless [0,1]).
$r_{\text{contingency,converter}}$	Contingency multiplier converter (dimensionless [0,1], if DC).
s_{soft}	Soft cost multiplier (decimal fraction). Default 0.10 (10%). Applied after contingency. Covers PM, engineering, A&G (MISO 2025b).
$g_{\text{construction}}$	Construction cost escalation rate (decimal). Default 2.5%/yr nominal (§A.2). Applied during delay.
C_{build}	Total build cost, nominal (\$). Terrain adjusted, contingency applied.
C_{build}	Total build cost (PV) (\$). Societal perspective; spending spread over construction and discounted at $r_{\text{WACC,real}}$.
T_{delay}	Delay years (years). From project technical details.
$T_{\text{construction}}$	Construction years (years). From project technical details.
t_{start}	Construction start year (year index). $t_{\text{start}} = T_{\text{delay}} + 1$ (1-based).
$r_{\text{WACC,real}}$	Real WACC (decimal). Societal discount rate (§A.2).
ξ_{DC}	Binary for DC projects (0 or 1). 1 if DC, 0 if AC.
$\xi_{\text{reconductoring}}$	Binary for reconductoring (0 or 1). 0 if reconductoring, 1 if not.

Variables

Equations

1. Component costs (raw miles).

$$C_{\text{conductor}} = V_{\text{conductor}} \cdot M_{\text{total}} + F_{\text{conductor}},$$

$$C_{\text{structure}} = V_{\text{structure}} \cdot M_{\text{total}} + F_{\text{structure}},$$

$$C_{\text{converter}} = V_{\text{converter}} \cdot M_{\text{total}} + F_{\text{converter}}.$$

In CTCC, $F_{\text{structure}} = 0$ and $V_{\text{converter}} = 0$.

2. Terrain-adjusted component costs.

$$C_{\text{adj,conductor}} = V_{\text{conductor}} \cdot M_{\text{weighted,total}} + F_{\text{conductor}},$$

$$C_{\text{adj,structure}} = V_{\text{structure}} \cdot M_{\text{weighted,total}},$$

$$C_{\text{adj,converter}} = F_{\text{converter}}.$$

3. Contingency-applied costs.

$$C_{\text{adj,cont,conductor}} = (1 + r_{\text{contingency,conductor}}) \cdot C_{\text{adj,conductor}},$$

$$C_{\text{adj,cont,structure}} = (1 + r_{\text{contingency,structure}}) \cdot C_{\text{adj,structure}},$$

$$C_{\text{adj,cont,converter}} = (1 + r_{\text{contingency,converter}}) \cdot C_{\text{adj,converter}}.$$

3b. Soft-cost-adjusted costs. A soft cost multiplier captures project management, engineering/design, and administrative and general costs not otherwise itemized:

$$C_{\text{adj,cont,conductor}} \times = (1 + s_{\text{soft}}), \quad C_{\text{adj,cont,structure}} \times = (1 + s_{\text{soft}}), \quad C_{\text{adj,cont,converter}} \times = (1 + s_{\text{soft}}).$$

The default $s_{\text{soft}} = 0.10$ (10%) is consistent with MISO MTEP planning methodology (MISO 2025b) and AACE 96R-18 owner's-cost/project-indirect-cost categories.

3c. Delay escalation (if $T_{\text{delay}} > 0$). Construction costs escalate during the delay period at the construction cost escalation rate, reflecting material, labor, and equipment cost growth between project conception and construction start (MISO 2025b):

$$C_{\text{adj,cont,conductor}} \times = (1 + g_{\text{construction}})^{T_{\text{delay}}},$$

$$C_{\text{adj,cont,structure}} \times = (1 + g_{\text{construction}})^{T_{\text{delay}}},$$

$$C_{\text{adj,cont,converter}} \times = (1 + g_{\text{construction}})^{T_{\text{delay}}}.$$

At the default $g_{\text{construction}} = 2.5\%/yr$ and a 5-year delay, construction costs increase by $(1.025)^5 = 1.13\times$ (13%). For SunZia's 17-year delay: $(1.025)^{17} = 1.52\times$ (52% cost increase).

4. Total build cost.

$$C_{\text{build}} = C_{\text{adj,cont,conductor}} + \xi_{\text{reconductoring}} \cdot C_{\text{adj,cont,structure}} + \xi_{\text{reconductoring}} \cdot \xi_{\text{DC}} \cdot C_{\text{adj,cont,converter}}.$$

Regulatory perspective (AFUDC). Build cost is AFUDC-eligible and is capitalized to COD using year-by-year S-curve compounding (see §A.3).

Societal perspective (present value). Build spending is spread evenly over $T_{\text{construction}}$ starting at $t_{\text{start}} = T_{\text{delay}} + 1$ and discounted at real WACC. For $T_{\text{construction}} > 0$:

$$C_{\text{build,annual}} = \frac{C_{\text{build}}}{T_{\text{construction}}}, \quad C_{\text{build}} = \sum_{t=t_{\text{start}}}^{t_{\text{start}}+T_{\text{construction}}-1} \frac{C_{\text{build,annual}}}{(1 + r_{\text{WACC,real}})^t}.$$

If $T_{\text{construction}} = 0$, the full cost is one-time at t_{start} : $C_{\text{build}} = C_{\text{build}}/(1 + r_{\text{WACC,real}})^{t_{\text{start}}}$.

B.1.2 Capital ROW costs (acquisition, holding)

This subsection defines capital ROW costs: one-time acquisition and annual holding (option fee) over the delay period. Only acquisition is AFUDC-eligible and enters rate base at COD; holding is an operating expense during the delay period and does not enter rate base. ROW rent is not a capital cost; it is an operational cost (§B.2.3). ROW here refers only to the linear corridor (miles $\times$ width by zone). Converter station acreage is not modeled.

Default calibration

Default acquisition and rent values ($p_{\text{acquisition,zone}}$, $p_{\text{rent,zone}}$) reflect the BLM Per-Acre Linear Rent Schedule for federal lands (Bureau of Land Management 2026b). These defaults apply to federal land; for private land, substitute state-level or appraised private-land values (e.g. USDA NASS county-level farmland values, state PUC filings, or project-specific appraisals). Per-zone acquisition and rent parameters may be varied in sensitivity analysis when project-specific land values differ from the BLM schedule.

Variable	Description
zone	Zone index (index). ROW split into zones (1–15). Use only zones where $M_{\text{zone}} > 0$.
M_{zone}	Miles in zone (miles).
W_{ROW}	ROW width (ft). Contingent on project category.
A_{zone}	Area of ROW of zone (acres).
$p_{\text{acquisition,zone}}$	Acquisition cost per acre, zone (\$/acre). One-time.
$p_{\text{hold,zone}}$	Holding (option fee) per acre per year, zone (\$/acre/year).
$p_{\text{rent,zone}}$	Rent per acre per year, zone (\$/acre/year). Operational only (§B.2.3).
$C_{\text{acquisition}}$	Total one-time acquisition cost (\$). From zone sum.
$C_{\text{hold,annual}}$	Total annual holding cost (\$/year). Permitting/delay years.
T_{delay}	Delay years (years). Used for holding.
$C_{\text{hold,total}}$	Total holding cost over delay (\$). Option fee over full delay period.
$\xi_{\text{acquisition}}$	Binary: acquisition counts toward capital ROW (0 or 1). 1 if acquisition applies, 0 if not (e.g. lease). Set by agreement type.
ξ_{holding}	Binary: holding counts toward capital ROW (0 or 1). 1 if holding applies, 0 if not. Set by agreement type.
$C_{\text{acquisition}}$	PV of acquisition (\$). Societal; discounted at $r_{\text{WACC,real}}$.
C_{hold}	PV of holding (\$). Societal; discounted at $r_{\text{WACC,real}}$.
$r_{\text{WACC,real}}$	Real WACC (decimal). Societal discount rate (§A.2).
$C_{\text{ROW,capital,nominal}}$	Total ROW capital costs, nominal (\$).
$C_{\text{ROW,capital}}$	Total ROW capital costs (PV) (\$). Societal perspective.

Agreement type	Acquisition	Holding
New Permanent Ease- ment	Yes ($\xi_{\text{acquisition}} = 1$)	Yes ($\xi_{\text{holding}} = 1$)
Simple Fee	Yes ($\xi_{\text{acquisition}} = 1$)	Yes ($\xi_{\text{holding}} = 1$)
Existing Lease/License	No ($\xi_{\text{acquisition}} = 0$)	No ($\xi_{\text{holding}} = 0$)
Federal Hybrid	Yes ($\xi_{\text{acquisition}} = 1$)	Yes ($\xi_{\text{holding}} = 1$)

Variables

Equations

1. Area of ROW per zone.

$$A_{\text{zone}} = \frac{M_{\text{zone}} \times 5280 \text{ (ft/mi)} \times W_{\text{ROW}}}{43\,560 \text{ (ft}^2\text{/acre)}}.$$

2. Total acquisition cost.

$$C_{\text{acquisition}} = \sum_{\text{zone}} A_{\text{zone}} p_{\text{acquisition,zone}}.$$

3. Annual holding cost.

$$C_{\text{hold,annual}} = \sum_{\text{zone}} A_{\text{zone}} p_{\text{hold,zone}}.$$

Default hold costs are set to 10% of annual rent per zone, approximating the option/reservation fee ratio observed in BLM and private ROW agreements.

4. Total holding cost over delay.

$$C_{\text{hold,total}} = C_{\text{hold,annual}} \times T_{\text{delay}}.$$

5. Total nominal ROW capital cost.

$$C_{\text{ROW,capital,nominal}} = \xi_{\text{acquisition}} \cdot C_{\text{acquisition}} + \xi_{\text{holding}} \cdot C_{\text{hold,total}}.$$

Regulatory perspective (AFUDC). ROW acquisition is AFUDC-eligible and is capitalized to COD using year-by-year S-curve compounding (see §A.3); holding costs are operating expenses during the delay period and do not enter rate base.

Societal perspective (present value). Discount at real WACC:

$$C_{\text{acquisition}} = \frac{C_{\text{acquisition}}}{(1 + r_{\text{WACC,real}})^{T_{\text{delay}}}}, \quad C_{\text{hold}} = \sum_{t=1}^{T_{\text{delay}}} \frac{C_{\text{hold,annual}}}{(1 + r_{\text{WACC,real}})^t}.$$

Total real (present) value of ROW capital costs:

$$C_{\text{ROW,capital}} = \xi_{\text{acquisition}} \cdot C_{\text{acquisition}} + \xi_{\text{holding}} \cdot C_{\text{hold}}.$$

B.1.3 Environmental mitigation

This subsection defines environmental mitigation costs: base costs (construction/restoration per effective acre by terrain and construction type) and credit costs (wetland and habitat off-site mitigation). The habitat set is the set of habitat terrain types: Forested, Scrubbed Flat, Desert/Barren, Rolling Hills, Mountain. The wetland set is the set of wetland terrain types: Wetland. Only habitat terrains with $M_{\text{terrain}} > 0$ and wetland terrains with $M_{\text{terrain}} > 0$ are used. For reconductoring projects, wetland and habitat credits are set to zero. Wetland credits use a single per-acre rate. Habitat credits use per-terrain rates.

Variable	Description
M_{terrain}	Miles of a specific terrain type (miles). Terrain set as in §A.1. Only terrains with $M_{\text{terrain}} > 0$ used.
W_{ROW}	ROW width (ft). Contingent on project category.
A_{terrain}	Base ROW area in terrain (acres). $A_{\text{terrain}} = (M_{\text{terrain}} \times 5280 \text{ (ft/mi)} \times W_{\text{ROW}}) / 43\,560 \text{ (ft}^2\text{/acre)}$.
$u_{\text{mitigation}}$	Mitigation uplift factor (dimensionless). Dimensionless (e.g. 1.25). Construction width beyond ROW (TCE).
$A_{\text{terrain}}^{\text{effective}}$	Effective acres for mitigation in terrain (acres). $A_{\text{terrain}}^{\text{effective}} = A_{\text{terrain}} \times u_{\text{mitigation}}$.
CT	Construction type (index). From project category (overhead, underground direct buried, tunnel, subsea).
$c_{\text{base,peracre}}(\text{terrain}, CT)$	Base mitigation cost per acre (\$/acre). Depends on terrain and construction type.
$C_{\text{base,total}}$	Total base mitigation/restoration cost (\$). Sum over terrains ($M_{\text{terrain}} > 0$) of $c_{\text{base,peracre}} \times A_{\text{terrain}}^{\text{effective}}$.
$r_{\text{marine,env}}$	Marine environmental mitigation rate (decimal fraction). Default 0.02 (2% of subsea construction CAPEX) (Champlain Hudson Power Express n.d.; Ofgem 2024; MISO 2025b). Applies to subsea construction type only.
$C_{\text{subsea,CAPEX}}$	Subsea construction CAPEX (conductor + structure, excl. converter) (\$). From build costs (§B.1.1), terrain-adjusted and contingency-applied.
$C_{\text{marine,env}}$	Marine environmental mitigation cost (\$). $C_{\text{marine,env}} = r_{\text{marine,env}} \times C_{\text{subsea,CAPEX}}$. Zero when construction type is not subsea.
$A_{\text{wetland}}^{\text{effective impact}}$	Wetland impact acres (effective) (acres). Wetland terrain only: $A_{\text{wetland}} \times u_{\text{mitigation}}$. Use only when $M_{\text{terrain}} > 0$ for wetland.
$A_{\text{habitat}}^{\text{effective impact}}$	Habitat impact acres (effective), total (acres). Sum over habitat terrains $\times u_{\text{mitigation}}$. Habitat = forested, scrubbed flat, desert barren, rolling hills, mountain. Use only when $M_{\text{terrain}} > 0$ for habitat terrains.
$A_{\text{terrain,habitat}}^{\text{effective impact}}$	Effective acres habitat impacted, that terrain (acres). $A_{\text{terrain}} \times u_{\text{mitigation}} \times \mu$ for that terrain. Used in $C_{\text{credits,habitat}}$. Only when $M_{\text{terrain}} > 0$ for that terrain.
p_{wetland}	Wetland credit cost per acre (\$/acre). Default \$51,361/acre (MISO 2025b).
$p_{\text{habitat}}(\text{terrain})$	Habitat credit cost per acre (\$/acre). Per terrain; explicit value for each habitat terrain.
$C_{\text{credits,wetlands}}$	Wetland credit cost (\$). Use only when $M_{\text{terrain}} > 0$ for wetland.
$C_{\text{credits,habitat}}$	Habitat credit cost (\$). Sum over habitat terrains of $p_{\text{habitat}}(\text{terrain}) \times A_{\text{terrain,habitat}}^{\text{effective impact}}$. Use only when $M_{\text{terrain}} > 0$ for habitat terrains.
μ	Compensatory mitigation ratio (dimensionless, ≥ 1). Required credits per impacted acre. Default 2.0 per 33 C.F.R. § 332.3(f) (U.S. Army Corps of Engineers 2008); Category II preservation ratio.
$\xi_{\text{reconductoring}}$	Binary: not reconductoring (0 or 1). 1 if not reconductoring, 0 if reconductoring. When 0, wetland and habitat credits set to zero.
$C_{\text{credits,total}}$	Total credit cost, wetland + habitat (\$). After reconductoring rule.
T_{delay}	Delay years (years).
$T_{\text{construction}}$	Construction years (years).
t_{start}	Construction start year (year index). $t_{\text{start}} = T_{\text{delay}} + 1$ (1-based).
$r_{\text{WACC,real}}$	Real WACC (decimal). Societal discount rate (§A.2).
$C_{\text{env.mit,nominal}}$	Total environmental mitigation cost, nominal (\$).
$B_{\text{env.mit,annual}}$	Annual base for env. mitigation (\$/year). Over construction years (PV calculation).
C_{credits}	PV of credits (\$).
$C_{\text{env.mit}}$	Total environmental mitigation cost (PV) (\$).

Variables

Equations

1. Base area per terrain.

$$A_{\text{terrain}} = \frac{M_{\text{terrain}} \times 5280 \text{ (ft/mi)} \times W_{\text{ROW}}}{43\,560 \text{ (ft}^2\text{/acre)}}.$$

2. Effective acres (uplift for TCE).

$$A_{\text{terrain}}^{\text{effective}} = A_{\text{terrain}} \times u_{\text{mitigation}}.$$

3. Total base mitigation cost.

$$C_{\text{base,total}} = \sum_{\text{terrain}} c_{\text{base,peracre}}(\text{terrain}, CT) \times A_{\text{terrain}}^{\text{effective}}.$$

3b. Marine environmental mitigation (subsea only). Marine environmental mitigation is modeled as a percentage of subsea construction CAPEX rather than a per-acre rate, since no marine mitigation banking market exists and environmental costs for offshore corridors scale with project complexity and value:

$$C_{\text{marine,env}} = r_{\text{marine,env}} \times C_{\text{subsea,CAPEX}}, \quad C_{\text{marine,env}} = 0 \text{ if } CT \neq \text{subsea}.$$

The default $r_{\text{marine,env}} = 0.02$ is consistent with the CHPE Environmental Trust Fund obligation (1.95% of project cost) (Champlain Hudson Power Express n.d.), Ofgem interconnector development expenditure data (1.4–2.7% across North Sea Link, Viking Link, and NeuConnect) (Ofgem 2024), and MISO’s 3% terrestrial environmental/permitting benchmark (MISO 2025b). $C_{\text{marine,env}}$ is added to $C_{\text{base,total}}$ and spread over the construction period on the same schedule as other base mitigation costs.

4. Wetland impact acres (wetland terrain only).

$$A_{\text{wetland}}^{\text{effective impact}} = A_{\text{wetland}} \times u_{\text{mitigation}} \times \mu.$$

5. Habitat impact acres, total.

$$A_{\text{habitat}}^{\text{effective impact}} = \left(\sum_{\text{terrain} \in \text{habitat}} A_{\text{terrain}} \right) \times u_{\text{mitigation}} \times \mu.$$

6. Wetland credit cost.

$$C_{\text{credits,wetlands}} = p_{\text{wetland}} \times A_{\text{wetland}}^{\text{effective impact}}.$$

7. Habitat credit cost.

$$C_{\text{credits,habitat}} = \sum_{\text{terrain} \in \text{habitat}} p_{\text{habitat}}(\text{terrain}) \times A_{\text{terrain,habitat}}^{\text{effective impact}}.$$

8. Total credit cost.

$$C_{\text{credits,total}} = \xi_{\text{reconductoring}} \times (C_{\text{credits,wetlands}} + C_{\text{credits,habitat}}).$$

9. Total nominal environmental mitigation cost.

$$C_{\text{env.mit,nominal}} = C_{\text{base,total}} + C_{\text{marine,env}} + C_{\text{credits,total}}.$$

Regulatory perspective (AFUDC). Base mitigation and credits are AFUDC-eligible and are capitalized to COD using year-by-year S-curve compounding (see §A.3). Total capitalized environmental mitigation equals capitalized base plus capitalized credits.

Societal perspective (present value). Base mitigation spread evenly over $T_{\text{construction}}$ starting at t_{start} . For $T_{\text{construction}} > 0$:

$$B_{\text{env.mit,annual}} = \frac{C_{\text{base,total}} + C_{\text{marine,env}}}{T_{\text{construction}}}, \quad C_{\text{base}} = \sum_{t=t_{\text{start}}}^{t_{\text{start}}+T_{\text{construction}}-1} \frac{B_{\text{env.mit,annual}}}{(1+r_{\text{WACC,real}})^t}.$$

If $T_{\text{construction}} = 0$, both base and credits are one-time at t_{start} . Otherwise, credits are spread over the construction period (same convention as base costs and AFUDC timing):

$$C_{\text{credits}} = \frac{C_{\text{credits,total}}}{T_{\text{construction}}} \sum_{t=t_{\text{start}}}^{t_{\text{start}}+T_{\text{construction}}-1} \frac{1}{(1+r_{\text{WACC,real}})^t}, \quad C_{\text{env.mit}} = C_{\text{base}} + C_{\text{credits}}.$$

Hard cost total

The three hard-cost components above sum to the hard cost bucket. Hard costs are the one-time capital expenditures that enter rate base and, from the societal perspective, are discounted from construction spending.

$$C_{\text{hard}} = C_{\text{build}} + C_{\text{ROW,capital}} + C_{\text{env.mit}}.$$

Component PV equations: C_{build} (§B.1.1), $C_{\text{ROW,capital}}$ (§B.1.2), $C_{\text{env.mit}}$ (§B.1.3).

B.2 Soft Costs (C_{soft})

Soft costs are ongoing operational, energy-related, and time-value costs; none are AFUDC-eligible or in rate base. Three sub-aggregates: operational (O&M, insurance, ROW rent), energy (thermal loss), and delay (base delay, congestion delay, curtailment delay, displacement delay).

B.2.1 O&M

This subsection defines operations and maintenance cost: conductor, converter, structure, and vegetation management by terrain (terrain set as in §A.1), from commercial operation date over project lifetime.

Variable	Description
M_{total}	Total line length (miles). From physical details.
M_{terrain}	Miles in terrain type (miles). Terrain set as in §A.1. Only terrains with $M_{\text{terrain}} > 0$ used in sums.
T_{delay}	Delay years (years). Before construction.
$T_{\text{construction}}$	Construction years (years).
T_{lifetime}	Project lifetime (years). O&M accrues over this.
T_{COD}	Commercial operation date (year index). $T_{\text{COD}} = T_{\text{delay}} + T_{\text{construction}} + 1$ (1-based; first year of O&M).
$r_{\text{WACC,real}}$	Real WACC (decimal). Used to discount O&M (§A.2).
CT	Construction type (index). Overhead, Underground direct-buried, Underground tunnel, Subsea.
c_{base}	Base line O&M rate (\$/mile/year). Overhead only. \$19,090/mi/yr, 2025\$: CT Siting Council (2022) \$29,636 (2017–2021 mean, ~2019\$) escalated to 2025\$ minus vegetation management (2025\$).
$r_{\text{converter,O\&M}}$	Converter O&M rate (decimal fraction). Annual O&M as fraction of converter station CAPEX. LCC: 0.005; VSC: 0.007 (Van Eeckhout et al. 2022; Kim, Jang, et al. 2012; Universidad de Castilla-La Mancha 2023).
$K_{\text{converter}}$	Total converter station CAPEX (\$). Equal to the fixed converter-station capital cost component of build cost (§B.1.1).
$r_{\text{line,O\&M}}$	Line O&M rate (decimal fraction). Non-overhead only. Annual O&M as fraction of line CAPEX. UG direct-buried: 0.0015 (Centre for Infrastructure Engineering and Technology (CIET) and Powerlink Queensland 2023); tunnel: 0.004 (Institution of Engineering and Technology and Mott MacDonald 2025); subsea: 0.025 (NeoMarketData 2025).
K_{line}	Line CAPEX (excl. converters) (\$). $K_{\text{line}} = C_{\text{conductor,build}} + C_{\text{structure,build}}$ from §B.1.1.
ξ_{DC}	Binary for DC projects (0 or 1). 1 if DC, 0 if AC. Gates converter O&M (§B.1.1).
$c_{\text{veg}}(\text{terrain})$	Vegetation management (\$/mile/year). By terrain. Overhead only.
$C_{\text{O\&M,annual}}$	Total annual O&M (\$/year). $C_{\text{line,annual}} + C_{\text{converter,annual}}$.
$C_{\text{O\&M,nominal}}$	O&M cost (nominal) (\$). Undiscounted lifetime O&M.
$g_{\text{O\&M}}$	Real O&M escalation rate (decimal). Default 2.0% real/yr (Connecticut Siting Council 2023). Applied via growing annuity, same formula as wildfire (§B.3.1) and outage (§B.3.2) escalation.
$C_{\text{O\&M}}$	O&M cost (PV) (\$). Present value of O&M from T_{COD} over T_{lifetime} .

Variables

Equations

1. COD (first year of O&M).

$$T_{\text{COD}} = T_{\text{delay}} + T_{\text{construction}} + 1.$$

2. Line O&M (annual).

$$C_{\text{line,annual}} = \begin{cases} c_{\text{base}} \times M_{\text{total}} + C_{\text{veg,annual}} & \text{if overhead} \\ r_{\text{line,O\&M}} \times K_{\text{line}} & \text{if non-overhead} \end{cases}$$

where:

$$C_{\text{veg,annual}} = \sum_{\text{terrain}} M_{\text{terrain}} \times c_{\text{veg}}(\text{terrain}).$$

The overhead base rate $c_{\text{base}} = \$19,090/\text{circuit-mile/year}$ (2025\$) is derived from the Connecticut Siting Council (2022) total O&M of \$29,636/circuit-mile (Connecticut Siting Council 2023)—based on FERC Form 1 data (2017–2021, ~2019\$)—escalated to 2025\$ at 2%/yr CAGR ($\times 1.126 \rightarrow \$33,370$), minus separately modeled vegetation management (\$14,367/mi forested default, 2025\$; U.S. Department of Energy 2024). This single rate covers conductor inspection, structure maintenance, patrol, and all non-vegetation line operating costs.

For non-overhead construction types, O&M scales with installed asset value rather than route length. No published source supports a conductor/structure decomposition for underground or subsea systems; all benchmarks report total O&M per circuit-mile or as a percentage of capital cost. Rates: underground direct-buried 0.15% (Centre for Infrastructure Engineering and Technology (CIET) and Powerlink Queensland 2023), consistent with Connecticut Siting Council FERC Form 1 data (\$22,937/circuit-mile; Connecticut Siting Council 2023); tunnel 0.4% (Institution of Engineering and Technology and Mott MacDonald 2025); subsea 2.5% (NeoMarketData 2025), consistent with Ofgem post-construction review data for Viking Link, North Sea Link, and IFA2 interconnectors.

Vegetation management defaults (overhead). Vegetation management costs $c_{\text{veg}}(\text{terrain})$ reflect U.S. utility spending data by terrain archetype, expressed in 2025\$ via CPI-U escalation from source years. Default values are calibrated to regulatory filings and industry publications: forested terrain (\$14,367/mi/yr) from Duke Energy Florida three-year cycle clearing costs (U.S. Department of Energy 2024); mountain terrain (\$12,314/mi/yr) reflecting helicopter access premiums of 5.7–8 $\times$ ground crew costs (Bonneville Power Administration 2021; LIFE Elia-RTE 2015); wetland (\$8,210/mi/yr) from Xcel Energy Minnesota data with regulatory access constraints (U.S. Department of Energy 2024); scrubbed/brush (\$5,131/mi/yr) from mechanical mulching costs

below forest clearing (Backwell 2024); rolling hills (\$3,850/mi/yr) from Duke Indiana IURC filing; farmland (\$1,196/mi/yr) and desert/barren (\$449/mi/yr) from AEP Texas data (Brown and Quanta Technology 2009); urban (\$1,000/mi/yr, author judgment: overhead transmission through dense urban cores is structurally rare and minimal vegetation exists where it does; no utility source isolates urban-specific transmission VM). Defaults may be varied for project-specific conditions including wildfire risk zone and local growth rates. Non-overhead construction types have zero vegetation management cost.

3. Converter (annual; DC only).

$$C_{\text{converter,annual}} = \xi_{\text{DC}} \times r_{\text{converter,O\&M}} \times K_{\text{converter}}.$$

Converter O&M scales with station capacity, not line length. The rate is technology-specific: 0.5% for LCC (thyristor-based, air-cooled) and 0.7% for VSC (IGBT submodules, water-cooled) per the literature consensus (Van Eeckhout et al. 2022; Kim, Jang, et al. 2012; Jiajie et al. 2019; Universidad de Castilla-La Mancha 2023).

4. Total annual O&M.

$$C_{\text{O\&M,annual}} = C_{\text{line,annual}} + C_{\text{converter,annual}}.$$

5. Nominal (undiscounted) lifetime O&M. O&M costs escalate at real rate $g_{\text{O\&M}}$ per year:

$$C_{\text{O\&M,nominal}} = \sum_{t=0}^{T_{\text{lifetime}}-1} C_{\text{O\&M,annual}} \times (1 + g_{\text{O\&M}})^t.$$

6. Real (present value) O&M. O&M costs escalate at real rate $g_{\text{O\&M}}$ from T_{COD} , using the same growing annuity formula as wildfire and outage risk costs:

$$C_{\text{O\&M}} = \frac{C_{\text{O\&M,annual}} \times \frac{1 - \left(\frac{1 + g_{\text{O\&M}}}{1 + r_{\text{WACC,real}}} \right)^{T_{\text{lifetime}}}}{r_{\text{WACC,real}} - g_{\text{O\&M}}}}{(1 + r_{\text{WACC,real}})^{T_{\text{delay}} + T_{\text{construction}}}}.$$

When $g_{\text{O\&M}} = 0$, this reduces to the flat-annuity form. The default $g_{\text{O\&M}} = 2.0\%$ real/yr is the Connecticut Siting Council's official 40-year lifecycle compound annual growth rate for transmission O&M (Connecticut Siting Council 2023), cross-validated against LBNL FERC Form 1 data (0.5–2.0% real CAGR across 117 investor-owned utilities, 2021) (LBNL 2021).

B.2.2 Operational insurance

This subsection defines operational risk-bearing cost: annual premiums (or self-insurance reserves) on insurable asset value from COD to end of life. Overhead and underground lines are predominantly self-insured (commercial property insurance for T&D lines is generally unavailable); subsea cables and converter stations are commercially insured. The rate $r_{\text{premium}}(CT)$ represents total annualized risk-bearing cost regardless of mechanism.

Variable	Description
$C_{\text{adj,cont,conductor}}$	Conductor build cost (\$). Terrain-adjusted, contingency applied. From Build costs (§B.1.1).
$C_{\text{adj,cont,structure}}$	Structure build cost (\$). Terrain-adjusted, contingency applied. From Build costs (§B.1.1).
$C_{\text{adj,cont,converter}}$	Converter build cost (\$). Terrain-adjusted, contingency applied. From Build costs (§B.1.1).
ξ_{DC}	Binary for DC projects (0 or 1). 1 if DC, 0 if AC. Gates converter in insurable value (§B.1.1).
$\xi_{\text{ins,conductor}}$	Include conductors in insurable value (0 or 1). 1 if insured, 0 if not.
$\xi_{\text{ins,structure}}$	Include structures in insurable value (0 or 1). 1 if insured, 0 if not.
$\xi_{\text{ins,converter}}$	Include converters in insurable value (0 or 1). 1 if insured, 0 if not.
$V_{\text{insurable}}$	Insurable asset value (\$). Sum of included build-cost components (with ξ_{DC} for converter).
CT	Construction type (index). Overhead, Underground direct-buried, Underground tunnel, Subsea.
$r_{\text{premium}}(CT)$	Premium rate (decimal). Annual premium as fraction of insurable value. Can depend on CT .
$r_{\text{premium,default}}$	Default premium rate (decimal). Used when no type-specific rate is defined for CT .
$C_{\text{insurance,annual}}$	Annual operational insurance premium (\$/year).
T_{lifetime}	Project lifetime (years).
T_{COD}	Commercial operation date (year index). First year premium is paid. $T_{\text{COD}} = T_{\text{delay}} + T_{\text{construction}} + 1$.
$r_{\text{WACC,real}}$	Real WACC (decimal). Used to discount premiums (§A.2).
$C_{\text{insurance,nominal}}$	Operational insurance cost (nominal) (\$). Undiscounted sum of premiums over project lifetime.
g_{ins}	Insurance escalation rate (decimal). Default 2.0% real/yr. Reflects structural climate risk repricing (WTW 2025; Powerlink Queensland 2025).
$C_{\text{insurance}}$	Operational insurance cost (PV) (\$). Present value of premium stream from T_{COD} over T_{lifetime} .

Variables

Equations

1. Insurable asset value.

$$V_{\text{insurable}} = \xi_{\text{ins,conductor}} \cdot C_{\text{adj,cont,conductor}} + \xi_{\text{ins,structure}} \cdot C_{\text{adj,cont,structure}} + \xi_{\text{ins,converter}} \cdot \xi_{\text{DC}} \cdot C_{\text{adj,cont,converter}}$$

The insurable value includes contingency and soft-cost multiplier applied upstream in §B.1.1.

2. Premium rate.

$$r_{\text{premium}}(CT) = \begin{cases} r_{\text{premium},CT} & \text{if a rate is specified for construction type } CT \\ r_{\text{premium,default}} & \text{otherwise} \end{cases}$$

3. Annual premium. The premium is computed separately for line assets and converter stations, since converter stations carry higher commercial rates:

$$C_{\text{insurance,annual}} = V_{\text{insurable,line}} \times r_{\text{premium,line}}(CT) + V_{\text{insurable,conv}} \times r_{\text{premium,conv}},$$

where $V_{\text{insurable,line}} = \xi_{\text{ins,conductor}} \cdot C_{\text{adj,cont,conductor}} + \xi_{\text{ins,structure}} \cdot C_{\text{adj,cont,structure}}$ and $V_{\text{insurable,conv}} = \xi_{\text{ins,converter}} \cdot \xi_{\text{DC}} \cdot C_{\text{adj,cont,converter}}$.

4. Nominal (undiscounted) lifetime cost.

$$C_{\text{insurance,nominal}} = C_{\text{insurance,annual}} \times \frac{(1 + g_{\text{ins}})^{T_{\text{lifetime}}} - 1}{g_{\text{ins}}}.$$

5. Real (present value) cost. Risk-bearing costs escalate at real rate g_{ins} from T_{COD} :

$$C_{\text{insurance}} = \frac{C_{\text{insurance,annual}} \times \frac{1 - \left(\frac{1 + g_{\text{ins}}}{1 + r_{\text{WACC,real}}} \right)^{T_{\text{lifetime}}}}{r_{\text{WACC,real}} - g_{\text{ins}}}}{(1 + r_{\text{WACC,real}})^{T_{\text{delay}} + T_{\text{construction}}}}.$$

Default calibration

Construction type	$r_{\text{premium}}(CT)$	Basis
Overhead	0.20%	Self-insurance reserve (National Association of Regulatory Utility Commissioners 2024; Florida Power & Light 2026; Powerlink Queensland 2025)
Underground (direct-buried)	0.20%	Same mechanism
Underground (tunnel)	0.20%	Same mechanism
Subsea	3.0%	Commercial PD-only (MarketIntel 2025; Ofgem 2024)
Converter	0.70%	Commercial (mGrid 2026; Ofgem 2024)

B.2.3 Operational ROW rent

This subsection defines operational ROW rent: annual payment for use of the right-of-way in zones where the line is built. Default per-acre zone values and annual rents from the BLM Per-Acre Linear Rent Schedule for Calendar Years 2026–2035 (Bureau of Land Management 2026b); these defaults apply to federal (BLM) lands and private-land projects should substitute site-specific values. Same zones, areas, and per-acre rent as in Capital ROW (§B.1.2); agreement type determines whether rent applies and over which years.

Variable	Description
zone	Zone index (index). Same zones as Capital ROW; use only zones where $M_{\text{zone}} > 0$.
M_{zone}	Miles in zone (miles).
W_{ROW}	ROW width (ft). Same as Capital ROW (project category).
A_{zone}	Area of ROW of zone (acres). $A_{\text{zone}} = (M_{\text{zone}} \times 5280 \text{ (ft/mi)} \times W_{\text{ROW}}) / 43\,560 \text{ (ft}^2\text{/acre)}$.
$p_{\text{rent,zone}}$	Rent per acre per year, zone (\$/acre/year).
$C_{\text{rent,annual}}$	Total annual ROW rent (\$/year). Sum over zones.
ξ_{rent}	Binary: rent applies (0 or 1). 1 for Existing Lease/License or Federal Hybrid; 0 for New Permanent Easement or Simple Fee.
T_{delay}	Delay years (years).
$T_{\text{construction}}$	Construction years (years).
T_{lifetime}	Project lifetime (years). Years of operation.
$t_{\text{rent,start}}$	First year rent is paid (year index). Agreement-dependent (see table).
T_{rent}	Number of years rent is paid (years). Agreement-dependent (see table).
g_{rent}	ROW rent escalation rate (decimal). Real annual escalation applied to annual rent. Default 0% real (§A.2).
$r_{\text{WACC,real}}$	Real WACC (decimal). Used to discount rent (§A.2).
$C_{\text{rent,total,nominal}}$	ROW rent cost (nominal) (\$). Undiscounted sum over all years rent is paid.
$C_{\text{rent,total}}$	ROW rent cost (PV) (\$). Present value of rent stream.

Agreement type	Rent?	$t_{\text{rent,start}}$	T_{rent}
New Permanent Easement	No ($\xi_{\text{rent}} = 0$)	—	—
Simple Fee	No ($\xi_{\text{rent}} = 0$)	—	—
Existing Lease/License	Yes ($\xi_{\text{rent}} = 1$)	1	$T_{\text{delay}} + T_{\text{construction}} + T_{\text{lifetime}}$
Federal Hybrid	Yes ($\xi_{\text{rent}} = 1$)	$T_{\text{delay}} + 1$	$T_{\text{construction}} + T_{\text{lifetime}}$

Variables

Equations

1. Area of ROW per zone.

$$A_{\text{zone}} = \frac{M_{\text{zone}} \times 5280 \text{ (ft/mi)} \times W_{\text{ROW}}}{43\,560 \text{ (ft}^2\text{/acre)}}.$$

2. Annual ROW rent.

$$C_{\text{rent,annual}} = \sum_{\text{zone}} A_{\text{zone}} p_{\text{rent,zone}}.$$

3. Nominal (undiscounted) ROW rent cost. Growing annual rent at rate g_{rent} over T_{rent} years:

$$C_{\text{rent,total,nominal}} = \xi_{\text{rent}} \times C_{\text{rent,annual}} \times \frac{(1 + g_{\text{rent}})^{T_{\text{rent}}} - 1}{g_{\text{rent}}}.$$

At $g_{\text{rent}} = 0$ this reduces to $\xi_{\text{rent}} \times C_{\text{rent,annual}} \times T_{\text{rent}}$.

4. Real (present value) ROW rent cost. Growing annuity from $t_{\text{rent,start}}$ for T_{rent} years, discounted at $r_{\text{WACC,real}}$:

$$C_{\text{rent,total}} = \frac{\xi_{\text{rent}} \times C_{\text{rent,annual}} \times \frac{1 - \left(\frac{1 + g_{\text{rent}}}{1 + r_{\text{WACC,real}}}\right)^{T_{\text{rent}}}}{r_{\text{WACC,real}} - g_{\text{rent}}}}{(1 + r_{\text{WACC,real}})^{t_{\text{rent,start}} - 1}}.$$

At $g_{\text{rent}} = 0$ this reduces to the flat-annuity form. The structure matches the growing-annuity PV in §B.3.1.

The default $g_{\text{rent}} = 0$ (0% real) reflects that federal BLM rents adjust at the Implicit Price Deflator for GDP rate (2.6%/yr nominal for CY 2026–2035), approximately matching inflation (Bureau of Land Management 2026a; U.S. Government 2024). State and private ROW leases similarly escalate at 2.5–3%/yr nominal—no evidence supports systematic above-inflation rent growth for transmission corridors. A positive g_{rent} may be examined in sensitivity analysis for urban corridor appreciation scenarios.

B.2.4 Energy losses

This subsection defines energy losses: the physical quantity of energy lost on the transmission path (line and, for DC, converters), used for energy loss costs (§B.2.5) and emissions costs (§B.4).

Variable	Description
C_{new}	New line capacity (MW). Nameplate. Reconductoring and greenfield.
V	Line voltage (kV). Reconductoring: existing (old) config; greenfield: new config.
L	Line length (miles). Total line length.
u	Line utilization (dimensionless). $0 \leq u \leq 1$. Average fraction of capacity used.
n_{phases}	Number of phases (count). 3 for AC; 2 for DC (poles).
n_{circuits}	Number of circuits / poles (count). Parallel circuits (AC) or poles (DC).
$n_{\text{conductorsperphase}}$	Conductors per phase (or per pole) (count). Conductors in parallel per phase/pole.
R_{mi}	Resistance per mile (Ω/mile). AC: 75°C; DC: corrected to operating temperature. Depends on conductor and circuit.
α_{20}	Temperature coefficient of resistance at 20°C (K^{-1}). DC only. Aluminum: 0.00403; copper: 0.00393 (IEC 2021).
$T_{\text{DC,op}}$	DC operating temperature (°C). Default 75°C, matching the AC 75°C convention (IEEE 2023; IEC 2023).
I	Phase current (A). RMS phase current at full capacity and nominal voltage.
ϕ_{AC}	AC power factor (dimensionless). e.g. 0.95. AC only.
f_{load}	Full-load adjustment factor (dimensionless). $0.3u + 0.7u^2$; empirical coefficients for North American transmission (NaitBelaid, Heleno, and LaCommare 2024).
H	Hours per year (h/year). e.g. 8760.
T_{lifetime}	Project lifetime (years).
ξ_{DC}	Binary for DC (0 or 1). 1 if DC, 0 if AC.
N_{conv}	Number of converter stations (count). 0 for AC; typically 2 for DC.
λ_{conv}	Converter loss fraction (dimensionless). Fraction of through-power lost per station (e.g. LCC, VSC).
$P_{\text{lineloss,MW}}$	Line loss (power) (MW). Total resistive line loss (average power).
$E_{\text{lineloss,MW,annual}}$	Line energy losses (annual) (MWh/year).
$E_{\text{lineloss,lifetime}}$	Line energy losses (lifetime) (MWh).
$P_{\text{corona,MW}}$	Corona loss (power) (MW). AC overhead only; voltage-dependent, load-independent. Zero for DC and non-overhead.
$c_{\text{corona}}(V)$	Corona loss by voltage class (kW/mile). 0 (≤ 345 kV), 6 (500 kV), 25 (765 kV); annual average, sea level (American Electric Power n.d.; Lawrence Berkeley National Laboratory 2014).
$P_{\text{converterloss,MW}}$	Converter loss (power) (MW). DC only.
$E_{\text{converterloss,MW,annual}}$	Converter energy losses (annual) (MWh/year).
E_{loss}	Total energy losses (annual) (MWh/year). Line + converter.
$E_{\text{totalloss,lifetime}}$	Total energy losses (lifetime) (MWh).

Variables

Equations

1. Phase current (from capacity and voltage).

$$I = \frac{C_{\text{new}} \times 1000 \text{ (A}\cdot\text{kV/MW)}}{\phi_{\text{AC}} \times V \times \sqrt{n_{\text{phases}}} \times n_{\text{circuits}}} \quad (\text{AC}), \quad I = \frac{C_{\text{new}} \times 1000 \text{ (A}\cdot\text{kV/MW)}}{V \times \sqrt{n_{\text{phases}}} \times n_{\text{circuits}}} \quad (\text{DC}).$$

2. Full-load adjustment (utilization).

$$f_{\text{load}} = 0.3u + 0.7u^2.$$

3. DC resistance temperature correction (DC only). DC resistance is tabulated at 20°C reference temperature but conductors operate at elevated temperature under load. Per IEC 60287-1-1:2023 and IEEE Std 738-2023, resistance is corrected linearly:

$$R_{\text{mi}} = R_{\text{DC},20} \times [1 + \alpha_{20}(T_{\text{DC,op}} - 20)] \quad (\text{DC only}).$$

At the default $T_{\text{DC,op}} = 75^\circ\text{C}$, this increases DC resistance by 22.2% for aluminum ($\alpha_{20} = 0.00403$) and 21.6% for copper ($\alpha_{20} = 0.00393$) relative to the uncorrected 20°C value (IEC 2021). AC resistance ($R_{\text{mi}} = R_{\text{AC},75}$) is already tabulated at 75°C and requires no correction.

4. Line loss (power). Total resistive (I^2R) line loss, average power in MW:

$$P_{\text{lineloss,MW}} = \left(\frac{I}{n_{\text{conductorsperphase}}} \right)^2 \times (n_{\text{conductorsperphase}} \times n_{\text{phases}} \times n_{\text{circuits}}) \times R_{\text{mi}} \times L \times \frac{f_{\text{load}}}{10^6}.$$

5. Line energy losses (annual and lifetime).

$$E_{\text{lineloss,MW,annual}} = P_{\text{lineloss,MW}} \times H, \quad E_{\text{lineloss,lifetime}} = E_{\text{lineloss,MW,annual}} \times T_{\text{lifetime}}.$$

6. Corona loss (AC overhead only). Corona loss occurs when conductor-surface electric field strength exceeds the dielectric breakdown gradient of air; it is voltage-dependent and load-independent, so it is modeled as an additive term rather than a multiplier on I^2R losses:

$$P_{\text{corona,MW}} = \xi_{\text{AC}} \times \xi_{\text{overhead}} \times \frac{c_{\text{corona}}(V) \times L \times n_{\text{circuits}}}{1000}.$$

Corona is added to total line loss before computing annual and lifetime energy losses: $P_{\text{lineloss,MW}} += P_{\text{corona,MW}}$. Values are annual averages at sea level, weighted by a typical 20%/2%/78% rain/snow/fair-weather mix (American Electric Power n.d.); higher altitude increases corona by approximately 25% per 1,000 m of elevation, and foul-weather corona can exceed fair-weather corona by 20:1. DC corona exists but is much smaller than AC corona and is excluded.

7. Converter loss (DC only).

$$P_{\text{converterloss,MW}} = \xi_{\text{DC}} \times N_{\text{conv}} \times \lambda_{\text{conv}} \times u \times C_{\text{new}}, \quad E_{\text{converterloss,MW,annual}} = P_{\text{converterloss,MW}} \times H.$$

Default converter loss fractions: LCC 0.75%, VSC 1.0%. The VSC default reflects a midpoint of the published range: ACORE reports 0.7% for modular multi-level converters (American Council on Renewable Energy 2021), while broader VSC survey data spans 1.2–1.5% (E3S Web of Conferences 2020).

8. Total energy losses.

$$E_{\text{loss}} = E_{\text{lineloss,MW,annual}} + E_{\text{converterloss,MW,annual}}, \quad E_{\text{totalloss,lifetime}} = E_{\text{loss}} \times T_{\text{lifetime}}.$$

B.2.5 Thermal loss costs

This subsection defines thermal loss cost: the cost of the energy lost on the transmission path (line and, for DC, converters), valuing energy losses from §B.2.4 at the value of load—the demand-side marginal value of the energy that would have been delivered. Per-conductor AC resistance at 75°C is sourced from the Southwire Overhead Conductor Manual (2018) for ACSR configurations, CTC Global published datasheets for ACCC, and IEC 60228/60287 for underground XLPE cables.

Variable	Description
E_{loss}	Total energy losses (annual) (MWh/year). From §B.2.4 (line + converter).
$E_{\text{lineloss,MW,annual}}$	Line energy losses (annual) (MWh/year). Conductor only.
$E_{\text{converterloss,MW,annual}}$	Converter energy losses (annual) (MWh/year). DC only.
v_{load}	Value of load (\$/MWh). Marginal value of energy to end-use load. The value of electricity derives from the services load performs with it, making demand-side valuation the correct welfare measure.
T_{lifetime}	Project lifetime (years).
T_{COD}	Commercial operation date (year). First year of operation.
$r_{\text{WACC,real}}$	Real WACC (decimal). Used to discount this cost (§A.2).
$C_{\text{lineloss,annual}}$	Line loss cost (annual) (\$/year). Cost of line losses only.
$C_{\text{converterloss,annual}}$	Converter loss cost (annual) (\$/year). Cost of converter losses only.
$C_{\text{loss,annual}}$	Total thermal loss cost (annual) (\$/year). Line + converter.
g_{benefit}	Benefit price escalation rate (decimal). From §A.2. Applied to energy valuation for welfare consistency.
$C_{\text{loss,nominal}}$	Total thermal loss cost (nominal) (\$). Undiscounted sum over lifetime.
C_{loss}	Total thermal loss cost (PV) (\$). PV of growing annual stream from T_{COD} over T_{lifetime} .

Variables

Equations

1. Component costs (line and converter).

$$C_{\text{lineloss,annual}} = E_{\text{lineloss,MW,annual}} \times v_{\text{load}}, \quad C_{\text{converterloss,annual}} = E_{\text{converterloss,MW,annual}} \times v_{\text{load}}.$$

2. Total annual thermal loss cost.

$$C_{\text{loss,annual}} = E_{\text{loss}} \times v_{\text{load}} = C_{\text{lineloss,annual}} + C_{\text{converterloss,annual}}.$$

3. Nominal (undiscounted) lifetime cost. Growing annual loss cost at rate g_{benefit} over T_{lifetime} years:

$$C_{\text{loss,nominal}} = C_{\text{loss,annual}} \times \frac{(1 + g_{\text{benefit}})^{T_{\text{lifetime}}} - 1}{g_{\text{benefit}}}.$$

At $g_{\text{benefit}} = 0$ this reduces to $C_{\text{loss,annual}} \times T_{\text{lifetime}}$.

4. Real (present value) cost. Growing annuity from T_{COD} for T_{lifetime} years, discounted at $r_{\text{WACC,real}}$:

$$C_{\text{loss}} = \frac{C_{\text{loss,annual}} \times \frac{1 - \left(\frac{1 + g_{\text{benefit}}}{1 + r_{\text{WACC,real}}} \right)^{T_{\text{lifetime}}}}{r_{\text{WACC,real}} - g_{\text{benefit}}}}{(1 + r_{\text{WACC,real}})^{T_{\text{delay}} + T_{\text{construction}}}}.$$

At $g_{\text{benefit}} = 0$ this reduces to the flat-annuity form. The structure matches the growing-annuity PV in §B.3.1.

The value of a MWh lost on the line is the same commodity as a MWh delivered to load (v_{load} , §C.2.1). Welfare consistency requires that both escalate at g_{benefit} . The loss-compensation emissions cost ($C_{\text{comp,emissions}}$) is separate and escalates at $g_{c,k}$ (SCC growth, §B.4).

Energy loss cost aggregate

Approach B’s single-constraint decomposition (§A.4) allocates all constrained hours to either congestion or curtailment benefit, so there is no separate residual-exceedance cost; the energy cost aggregate is thermal loss cost only:

$$C_{\text{energy}} = C_{\text{loss}}.$$

B.2.6 Base delay costs

This subsection defines base delay costs: the total direct out-of-pocket annual expenditure during the delay (permitting and pre-construction) period. This captures all annual project-sustaining costs—legal, administrative, labor, regulatory, project management, and similar—as a single exogenous annual cost parameter. No published source provides benchmark rates for these costs at the component level; they are project-specific. Foregone congestion and curtailment benefits during delay are modeled separately (§B.2.7, §B.2.8). Total nominal cost is the annual cost times the number of delay years. Present value is a level annuity over the delay period, discounted at real WACC.

Variable	Description
$C_{\text{delay,annual}}$	Annual base delay cost (\$/year). Total direct out-of-pocket annual spend during delay. Exogenous parameter; default 0.
T_{delay}	Delay years (years). From project technical details; permitting/pre-construction period.
$r_{\text{WACC,real}}$	Real WACC (decimal). Used to discount the delay cost stream (§A.2).
$C_{\text{delay,nominal}}$	Total base delay cost (nominal) (\$). $C_{\text{delay,nominal}} = C_{\text{delay,annual}} \times T_{\text{delay}}$.
$C_{\text{base,delay}}$	Total base delay cost (PV) (\$). PV of level annuity over T_{delay} years at $r_{\text{WACC,real}}$.

Variables

Equations

1. Annual base delay cost. $C_{\text{delay,annual}}$ is specified directly (no decomposition).

2. Nominal total base delay cost.

$$C_{\text{delay,nominal}} = C_{\text{delay,annual}} \times T_{\text{delay}}.$$

3. Real (present value) base delay cost. Level annuity over years $t = 1, \dots, T_{\text{delay}}$, discounted at $r_{\text{WACC,real}}$:

$$C_{\text{base,delay}} = C_{\text{delay,annual}} \times \frac{1 - (1 + r_{\text{WACC,real}})^{-T_{\text{delay}}}}{r_{\text{WACC,real}}}.$$

Link: Congestion delay costs (§B.2.7) and curtailment delay costs (§B.2.8) are opportunity costs of foregone congestion/curtailment relief during delay and construction; see those sections. All three delay cost categories apply only during the pre-COD period (delay and construction); the benefit streams in §C apply only post-COD. A given year falls in exactly one of these periods, so there is no double-count between delay costs and benefits.

B.2.7 Congestion delay costs

This subsection defines congestion delay costs: the opportunity cost of congestion that is not relieved while the project is in the delay and construction period. In each of those years, the same congestion relief (MWh/year) that the project would provide once in service is valued at the same price (\$/MWh), growing at g_{benefit} ; that annual value is the cost of delay. The annual congestion delay cost equals the annual congestion reduction benefit. The cost stream runs over $T_{\text{delay}} + T_{\text{construction}}$ years; present value is a growing annuity over that period, discounted at real WACC. Conceptually: this cost answers “What congestion benefit do we give up each year we are delayed?” Capacity relief and the congestion-fraction split (Approach B) are the same as in §C. The same annual \$ amount is used as the congestion benefit, so there is no double-count with post-COD benefits.

Variable	Description
$B_{\text{congestion,annual}}$	Annual congestion reduction benefit (\$/year). Same formula as in §C: allocated congestion relief MWh/year $\times \gamma_{\text{congestion}}$.
$C_{\text{cong,delay,annual}}$	Congestion delay cost (annual) (\$/year). $C_{\text{cong,delay,annual}} = B_{\text{congestion,annual}}$.
T_{delay}	Delay years. From project technical details.
$T_{\text{construction}}$	Construction years. From project technical details.
$r_{\text{WACC,real}}$	Real WACC (decimal). Used to discount the delay cost stream (§A.2).
g_{benefit}	Benefit price escalation rate (decimal). From §A.2. The same commodity price growth that applies to the post-COD congestion benefit.
$C_{\text{cong,delay,nominal}}$	Congestion delay cost (nominal) (\$). Total over delay + construction; undiscounted.
$C_{\text{cong,delay}}$	Congestion delay cost (PV) (\$). PV of growing annuity over $T_{\text{delay}} + T_{\text{construction}}$ years at $r_{\text{WACC,real}}$, starting at year 1.

Variables

Equations

1. Annual congestion delay cost. The annual opportunity cost equals the annual congestion reduction benefit (allocated congestion relief $\times \gamma_{\text{congestion}}$; see §C):

$$C_{\text{cong,delay,annual}} = B_{\text{congestion,annual}}.$$

2. Nominal total congestion delay cost. Growing annual cost at rate g_{benefit} over $T_{\text{delay}} + T_{\text{construction}}$ years:

$$C_{\text{cong,delay,nominal}} = C_{\text{cong,delay,annual}} \times \frac{(1 + g_{\text{benefit}})^{T_{\text{delay}} + T_{\text{construction}}} - 1}{g_{\text{benefit}}}.$$

At $g_{\text{benefit}} = 0$ this reduces to $C_{\text{cong,delay,annual}} \times (T_{\text{delay}} + T_{\text{construction}})$.

3. Real (present value) congestion delay cost. Growing annuity over years $t = 1, \dots, T_{\text{delay}} + T_{\text{construction}}$, discounted at $r_{\text{WACC,real}}$:

$$C_{\text{cong,delay}} = C_{\text{cong,delay,annual}} \times \frac{1 - \left(\frac{1 + g_{\text{benefit}}}{1 + r_{\text{WACC,real}}} \right)^{T_{\text{delay}} + T_{\text{construction}}}}{r_{\text{WACC,real}} - g_{\text{benefit}}}.$$

At $g_{\text{benefit}} = 0$ this reduces to the flat-annuity form.

Link: The same effective capacity relief and congestion-fraction split (Approach B) define $B_{\text{congestion,annual}}$ (§C). This subsection uses that annual value as the cost per year of delay/construction, so benefits (post-COD) and congestion delay cost (during delay/construction) are consistent and not double-counted.

Link: Curtailment delay costs (§B.2.8) and this subsection use the same period ($T_{\text{delay}} + T_{\text{construction}}$) and discounting ($r_{\text{WACC,real}}$). Both draw on the same congestion-fraction split, so the two delay costs are consistent and not double-counted.

Link: Base delay costs (§B.2.6) are out-of-pocket delay costs (legal, admin, etc.). Congestion delay costs are the opportunity cost of foregone congestion relief during the same period. Both use T_{delay} ; this subsection uses $T_{\text{delay}} + T_{\text{construction}}$ as the cost duration and $r_{\text{WACC,real}}$ for PV.

B.2.8 Curtailment delay costs

This subsection defines curtailment delay costs: the opportunity cost of curtailment that is not relieved while the project is in the delay and construction period. In each of those years, the same curtailment relief (MWh/year) that the project would provide once in service is valued at the same price (\$/MWh), growing at g_{benefit} ; that annual value is the cost of delay. The annual curtailment delay cost equals the annual curtailment reduction benefit. The cost stream runs over $T_{\text{delay}} + T_{\text{construction}}$ years; present value is a growing annuity over that period, discounted at real WACC. Conceptually: this cost answers “What curtailment benefit do we give up each year we are delayed?” Capacity relief and the congestion-fraction split (Approach B) are the same as in §C. The same annual \$ amount is used as the curtailment benefit, so there is no double-count with post-COD benefits.

Variable	Description
$B_{\text{curtailment,annual}}$	Annual curtailment reduction benefit (\$/year). Same formula as in §C: allocated curtailment relief MWh/year $\times \gamma_{\text{curtailment}}$.
$C_{\text{curt,delay,annual}}$	Curtailment delay cost (annual) (\$/year). $C_{\text{curt,delay,annual}} = B_{\text{curtailment,annual}}$.
T_{delay}	Delay years. From project technical details.
$T_{\text{construction}}$	Construction years. From project technical details.
$r_{\text{WACC,real}}$	Real WACC (decimal). Used to discount the delay cost stream (§A.2).
g_{benefit}	Benefit price escalation rate (decimal). From §A.2. The same commodity price growth that applies to the post-COD curtailment benefit.
$C_{\text{curt,delay,nominal}}$	Curtailment delay cost (nominal) (\$). Total over delay + construction; undiscounted.
$C_{\text{curt,delay}}$	Curtailment delay cost (PV) (\$). PV of growing annuity over $T_{\text{delay}} + T_{\text{construction}}$ years at $r_{\text{WACC,real}}$, starting at year 1.

Variables

Equations

1. Annual curtailment delay cost. The annual opportunity cost equals the annual curtailment reduction benefit (allocated curtailment relief $\times \gamma_{\text{curtailment}}$; see §C):

$$C_{\text{curt,delay,annual}} = B_{\text{curtailment,annual}}.$$

2. Nominal total curtailment delay cost. Growing annual cost at rate g_{benefit} over $T_{\text{delay}} + T_{\text{construction}}$ years:

$$C_{\text{curt,delay,nominal}} = C_{\text{curt,delay,annual}} \times \frac{(1 + g_{\text{benefit}})^{T_{\text{delay}} + T_{\text{construction}}} - 1}{g_{\text{benefit}}}.$$

At $g_{\text{benefit}} = 0$ this reduces to $C_{\text{curt,delay,annual}} \times (T_{\text{delay}} + T_{\text{construction}})$.

3. Real (present value) curtailment delay cost. Growing annuity over years $t = 1, \dots, T_{\text{delay}} + T_{\text{construction}}$, discounted at $r_{\text{WACC,real}}$:

$$C_{\text{curt,delay}} = C_{\text{curt,delay,annual}} \times \frac{1 - \left(\frac{1 + g_{\text{benefit}}}{1 + r_{\text{WACC,real}}} \right)^{T_{\text{delay}} + T_{\text{construction}}}}{r_{\text{WACC,real}} - g_{\text{benefit}}}.$$

At $g_{\text{benefit}} = 0$ this reduces to the flat-annuity form.

Link: The same effective capacity relief and congestion-fraction split (Approach B) define $B_{\text{curtailment,annual}}$ (§C). This subsection uses that annual value as the cost per year of delay/construction, so benefits (post-COD) and curtailment delay cost (during delay/construction) are consistent and not double-counted.

Link: Congestion delay costs (§B.2.7) and this subsection use the same period ($T_{\text{delay}} + T_{\text{construction}}$) and discounting ($r_{\text{WACC,real}}$). Both draw on the same congestion-fraction split, so the two delay costs are consistent and not double-counted.

Link: Base delay costs (§B.2.6) are out-of-pocket delay costs (legal, admin, etc.). Curtailment delay costs are the opportunity cost of foregone curtailment relief during the same delay and construction period.

Soft cost total

The nine soft-cost components above sum to the soft cost bucket. $C_{\text{emissions}}$ is a separate bucket (§B.4); see also §D. Soft costs are ongoing operational, energy-related, and time-value costs; none are AFUDC-eligible or in rate base.

$$C_{\text{operational}} = C_{\text{O\&M}} + C_{\text{insurance}} + C_{\text{rent,total}},$$

$$C_{\text{energy}} = C_{\text{loss}},$$

$$C_{\text{delay}} = C_{\text{base,delay}} + C_{\text{cong,delay}} + C_{\text{curt,delay}} + C_{\text{disp,delay}},$$

$$C_{\text{soft}} = C_{\text{operational}} + C_{\text{energy}} + C_{\text{delay}}.$$

Component PV equations: $C_{\text{O\&M}}$ (§B.2.1), $C_{\text{insurance}}$ (§B.2.2), $C_{\text{rent,total}}$ (§B.2.3), C_{loss} (§B.2.5), $C_{\text{base,delay}}$ (§B.2.6), $C_{\text{cong,delay}}$ (§B.2.7), $C_{\text{curt,delay}}$ (§B.2.8), $C_{\text{disp,delay}}$ (§C.3.1).

B.3 Risk Costs (C_{risk})

This subsection defines risk costs: expected cost of wildfires and expected cost of outages. None of these costs are AFUDC-eligible or included in rate base. These two sub-items produce the risk cost bucket C_{risk} (§B.3.2).

B.3.1 Expected cost of wildfires

This subsection defines expected cost of wildfires: the expected annual loss (EAL) from wildfire events over the project’s operating life. It uses a line-level base ignition rate scaled by construction type, multiplied by severity (\$ per event), with optional risk growth over the lifetime. Present value is a growing annuity from COD, typically discounted at the social discount rate.

Variable	Description
L	Total line length (miles). Sum of all terrain segment lengths from physical details.
f	Base ignition rate (events/(mi·yr)). Overhead baseline; line-level scalar.
k_t	Construction-type multiplier (dimensionless). Overhead 1.0; underground 0.01 (99% reduction; PG&E 2024); subsea 0.0 (no ignition risk).
f_t	Effective ignition rate for type t (events/(mi·yr)). $f_t = f \times k_t$.
λ_{wf}	Total annual event rate (events/year). $\lambda_{\text{wf}} = L \times f_t$.
$S_{\text{event, wf}}$	Severity (loss per event) (\$). Expected loss per wildfire event.
EAL_{wf}	Expected annual loss (\$/year). $EAL_{\text{wf}} = \lambda_{\text{wf}} \times S_{\text{event, wf}}$.
g_{wf}	Wildfire risk growth rate (decimal). Annual increase in expected loss (e.g. escalation of risk).
T_{lifetime}	Project lifetime (years).
T_{COD}	Commercial operation date (year index) (years). First year of operation; cost stream starts here.
T_{delay}	Delay years. Used in PV discounting.
$T_{\text{construction}}$	Construction years. Used in PV discounting.
r_{social}	Social discount rate (decimal). Used to discount this externality cost (§A.2).
$C_{\text{wf, nominal}}$	Expected wildfire cost (nominal) (\$). Sum of growing annual EAL_{wf} over lifetime; undiscounted.
C_{wf}	Expected wildfire cost (PV) (\$). PV of growing EAL_{wf} stream from T_{COD} over T_{lifetime} ; discounted for delay and construction.

Variables

Equations

1. Effective ignition rate (construction type).

$$f_t = f \times k_t.$$

2. Total annual event rate.

$$\lambda_{\text{wf}} = L \times f_t.$$

3. Expected annual loss.

$$EAL_{\text{wf}} = \lambda_{\text{wf}} \times S_{\text{event, wf}}.$$

4. Nominal (undiscounted) total cost. Growing annual loss at rate g_{wf} over T_{lifetime} years:

$$C_{\text{wf, nominal}} = EAL_{\text{wf}} \times \frac{(1 + g_{\text{wf}})^{T_{\text{lifetime}}} - 1}{g_{\text{wf}}}.$$

5. Real (present value) cost. Two reference dates: Year 0 = project analysis start (permitting, pre-construction); T_{COD} = commercial operation date = first year the cost stream begins ($T_{\text{delay}} + T_{\text{construction}}$ years after Year 0). Present value is expressed in Year-0 dollars.

Step 1: The numerator is the present value of the growing annuity of annual losses (EAL_{wf} growing at g_{wf}) over T_{lifetime} years, with “present” taken at T_{COD} . So it is the lump sum at T_{COD} equivalent to the stream from T_{COD} to end of life. Value at T_{COD} :

$$\text{PV at } T_{\text{COD}} = EAL_{\text{wf}} \times \frac{1 - \left(\frac{1 + g_{\text{wf}}}{1 + r_{\text{social}}} \right)^{T_{\text{lifetime}}}}{r_{\text{social}} - g_{\text{wf}}}.$$

Step 2: The denominator moves that lump sum from T_{COD} back to Year 0: the amount at T_{COD} is discounted by $(1 + r_{\text{social}})^{T_{\text{delay}} + T_{\text{construction}}}$.

The real (present value) cost is therefore:

$$C_{\text{wf}} = \frac{EAL_{\text{wf}} \times \frac{1 - \left(\frac{1 + g_{\text{wf}}}{1 + r_{\text{social}}} \right)^{T_{\text{lifetime}}}}{r_{\text{social}} - g_{\text{wf}}}}{(1 + r_{\text{social}})^{T_{\text{delay}} + T_{\text{construction}}}}.$$

Default calibration

The default base ignition rate $f = 0.002$ events/(mi-yr) is derived from CPUC-reportable transmission ignition data: PG&E’s Safety and Operational Metrics (SOM) Metric 3.16 reports a 9-year average (2015–2023) of 1.40 ignitions per 1,000 HFTD circuit-miles (Pacific Gas and Electric Company 2025), while SDG&E CPUC filing data yield 0.00244 fires/mi/yr for all-cause transmission ignitions (Johnson 2014). The 3:1 distribution-to-transmission ignition ratio (Taylor and Roald 2022) applied to the distribution rate from Warner, Callaway, and Fowlie (2025) produces a consistent cross-check of ~ 0.0017 /mi/yr. The default severity $S_{\text{event, wf}} = \$8,328,765$ per event (2025\$, CPI-U escalated from ~ 2023.5) is a probability-weighted average across fire size classes, anchored by the finding that fewer than 1% of grid-caused ignitions exceed 10,000 acres yet account for 79% of total acreage burned (Eagleston et al. 2025). Three independent cross-checks — PG&E total liability per ignition (\$6–7M), California annual property damage per ignition (\$4.5M), and a bottom-up size-class decomposition (\$13M) — bracket the recommended value. The risk growth rate $g_{\text{wf}} = 1.5\%$ /yr reflects a projected 45% cumulative increase in utility-ignited wildfire risk by 2050 (Warner, Callaway, and Fowlie 2025). Defaults are calibrated to fire-prone overhead transmission in the western United States; parameters may be varied in sensitivity analysis for other regions or risk profiles.

B.3.2 Expected cost of outages

This subsection defines expected cost of outages: the expected annual loss (EAL_{out}) from transmission outages over the project’s operating life. It uses a line-level outage rate by construction type, a base duration with construction-type multiplier, capacity at risk (fraction of line capacity lost per event), and value of lost load (VoLL) piecewise by duration. Present value is a growing annuity from COD, typically discounted at the social discount rate. The cost stream starts at COD (after delay and construction).

Variable	Description
CT	Construction type index. e.g. overhead, underground, subsea.
L	Total line length (miles). Sum of all terrain segment lengths from physical details.
r_{CT}	Outage rate for construction type CT (events/(mi-yr)). Underground/subsea: flat per-mile rate.
$\lambda_{\text{terminal}}$	Terminal outage rate (outages/yr). Overhead only. Default 0.3. Length-independent component (Landgren 1986; Adler et al. 1994).
$\lambda_{\text{per mile}}$	Per-mile outage rate (outages/(mi-yr)). Overhead only. Default 0.012. Exposure-dependent (Koval 1996; Pijarski and Belowski 2025).
λ_{out}	Total annual outage rate (events/year). Overhead: $\lambda_{\text{out}} = \lambda_{\text{terminal}} + \lambda_{\text{per mile}} \times L$. Underground/subsea: $\lambda_{\text{out}} = L \times r_{CT}$.
H_{base}	Base outage duration (h/event). Hours per outage event; line-level scalar.
k_{CT}	Duration multiplier for construction type CT (dimensionless). Overhead 1; underground, subsea can be > 1 .
H_{eff}	Effective outage duration (h/event). $H_{\text{eff}} = H_{\text{base}} \times k_{CT}$.
ϕ	Capacity-at-risk factor (dimensionless). $\phi = 1/N_{\text{poles}}$, where N_{poles} is the number of circuits (AC) or poles (DC) from the conductor specification. For single-circuit AC, $\phi = 1$; for bipole DC, $\phi = 0.5$.
C	Line capacity (MW). Project capacity (nameplate).
u	Line utilization factor (dimensionless). Average fraction of line capacity in use; from project technical details.
MW_{lost}	MW lost per event (MW). $MW_{\text{lost}} = \phi \times C \times u$.
ρ	Load-shed fraction (dimensionless). Fraction of lost flow that becomes unserved energy. AC meshed: $\rho = 0.05$; DC point-to-point: $\rho = 0.80$. Derived from AC/DC classification.
c_{redisp}	Redispatch cost (\$/MWh). Marginal congestion cost of rerouting power through alternative paths. Default \$24/MWh (2025\$).
U	Unserved energy per event (MWh/event). $U = H_{\text{eff}} \times \rho \times MW_{\text{lost}}$.
v_k	VoLL for duration tier k (\$/MWh). 10 tiers from 0–1 h through 720+ h; see VoLL tier table below.
$C_{\text{event, out}}$	Cost per outage event, \$/event. Sum of load-shed cost (piecewise VoLL on $\rho \cdot MW_{\text{lost}}$) and redispatch cost (c_{redisp} on $(1 - \rho) \cdot MW_{\text{lost}}$).
EAL_{out}	Expected annual loss (\$/year). $EAL_{\text{out}} = \lambda_{\text{out}} \times C_{\text{event, out}}$.
g_{outages}	Outage risk growth rate (decimal). Annual increase in expected cost (optional).
T_{lifetime}	Project lifetime (years).
T_{COD}	Commercial operation date (year index) (years). First year of operation; cost stream starts here.
T_{delay}	Delay years. Used in PV discounting.
$T_{\text{construction}}$	Construction years. Used in PV discounting.
r_{social}	Social discount rate (decimal). Used to discount this externality cost (§A.2).
$C_{\text{outage, nominal}}$	Expected outage cost (nominal) (\$). Sum of growing annual EAL_{out} over lifetime; undiscounted.
C_{outage}	Expected outage cost (PV) (\$). PV of growing EAL_{out} stream from T_{COD} over T_{lifetime} ; discounted for delay and construction.

Variables

Equations

1. Outage rate (by construction type). For overhead lines, the outage rate follows a decomposition model capturing length-independent terminal effects and exposure-dependent line failures:

$$\lambda_{\text{out}} = \lambda_{\text{terminal}} + \lambda_{\text{per mile}} \times L \quad (\text{overhead}).$$

For underground and subsea, the pure per-mile linear model applies:

$$\lambda_{\text{out}} = L \times r_{CT} \quad (\text{underground, subsea}).$$

Effective outage duration: $H_{\text{eff}} = H_{\text{base}} \times k_{CT}$.

The decomposition model (Landgren 1986) captures length-independent substation and protection equipment failures ($\lambda_{\text{terminal}}$, approximately 25% of total outages for ≥ 230 kV lines; Adler et al. 1994) separately from exposure-dependent conductor and insulator failures ($\lambda_{\text{per mile}}$, consistent with per-mile rates from CEA, BPA, and Polish TSO data; Koval 1996; Pijarski and Belowski 2025).

2. MW lost and unserved energy per event.

$$MW_{\text{lost}} = \phi \times C \times u, \quad U = H_{\text{eff}} \times \rho \times MW_{\text{lost}}.$$

3. Cost per event (piecewise VoLL). Duration is split into tiers with VoLL (\$/MWh) per tier. Tiers 1–5 are ERCOT system-wide survey values (Gibbons and Sergici 2024); tiers 6–9 are extrapolated; tier 10 is a flat cap at the 30-day value. Values are 2025\$ (CPI-U from Brattle/ERCOT 2024, $\times 1.0262$).

Tier	Duration	v_k (\$/MWh)	Source
1	0–1 h	36,620	Observed
2	1–2 h	21,885	Observed
3	2–4 h	13,689	Observed
4	4–8 h	10,708	Observed
5	8–16 h	13,937	Observed
6	16–32 h	20,628	Extrapolated
7	32–64 h	36,282	Extrapolated
8	64 h–7 d	88,512	Extrapolated
9	7–30 d	367,280	Extrapolated
10	30 d+	367,280	Cap

VoLL extrapolation model. Observed ERCOT system-wide VoLL (\$/MWh) declines sharply from 1 to 8 hours, then reverses at 16 hours. To extend beyond the 16-hour survey ceiling, a three-term model is fitted by ordinary least squares to the five observed values:

$$V(t) = \frac{a}{t} + b + ct, \quad a = 32,210, \quad b = 3,339, \quad c = 492.$$

The a/t term captures front-loaded disruption costs (production shutdown, restart); b is a steady-state floor; ct captures escalating societal costs for extended outages (food safety, medical, economic cascading). The model minimum occurs at $t^* = \sqrt{a/c} \approx 8.1$ hours, consistent with the observed trough at 8 hours. For $t > 720$ hours (30 days), VoLL is capped at \$367,280/MWh; outages beyond 30 days are assumed to transition from outage recovery to grid reconstruction, a regime outside the scope of this model.

Cost per event is the sum of a load-shed term (piecewise VoLL applied to the shed fraction) and a redispatch term (flat congestion cost applied to the rerouted fraction):

$$C_{\text{event,out}} = \underbrace{\sum_k (\text{hours in tier } k) \times \rho \cdot MW_{\text{lost}} \times v_k}_{\text{load-shed cost}} + \underbrace{(1 - \rho) \cdot MW_{\text{lost}} \cdot H_{\text{eff}} \cdot c_{\text{redisp}}}_{\text{redispatch cost}}.$$

Two-tier outage cost decomposition. In a well-planned meshed network, N-1 contingency criteria (NERC TPL-001-5.1) require that no single-element outage causes involuntary load shedding under normal conditions. Systems meeting the standard loss-of-load expectation of 0.1 days/year exhibit normalized expected unserved energy of only 1–10 parts per million of total demand. The economic consequence of most transmission outages is therefore not load shedding but congestion: the cost of redispatching more expensive generation to compensate for the lost transfer capability.

The outage cost per event is accordingly decomposed into two terms. The first term values the fraction ρ of lost flow that becomes actual unserved energy at the piecewise VoLL schedule. The second term values the remainder $(1 - \rho)$ at the marginal redispatch cost c_{redisp} , representing the congestion premium during the outage. This two-tier structure follows the ENTSO-E cost-benefit analysis framework (ENTSO-E 2021), which separately quantifies socio-economic welfare from congestion relief (indicator B1) and security-of-supply benefit from avoided load shedding (indicator B6), summing them without double-counting.

The load-shed fraction ρ is derived from AC/DC classification: $\rho = 0.05$ for AC lines in meshed networks, reflecting N-1 compliance and expected-unserved-energy standards; and $\rho = 0.80$ for DC point-to-point lines, reflecting limited alternative paths with partial reserve margin at the receiving end. The default redispatch cost $c_{\text{redisp}} = \$24/\text{MWh}$ (2025\$) is the midpoint of empirical congestion relief values from LBNL analysis of historical locational marginal prices across U.S. ISOs, which finds median values of \$11/MWh for regional links

and \$24/MWh for interregional links (Millstein et al. 2022). U.S. system-wide transmission congestion costs of \$6–21 billion annually (Shreve et al. 2025), across approximately 4,000 TWh of consumption, corroborate a system-average congestion cost of \$1.5–5/MWh.

4. Expected annual loss.

$$EAL_{\text{out}} = \lambda_{\text{out}} \times C_{\text{event,out}}.$$

5. Nominal (undiscounted) total cost.

Growing annual cost at rate g_{outages} over T_{lifetime} years:

$$C_{\text{outage,nominal}} = EAL_{\text{out}} \times \frac{(1 + g_{\text{outages}})^{T_{\text{lifetime}}} - 1}{g_{\text{outages}}}.$$

6. Real (present value) cost.

Two reference dates: Year 0 = project analysis start (permitting, pre-construction); T_{COD} = commercial operation date = first year of operation ($T_{\text{delay}} + T_{\text{construction}}$ years after Year 0). Present value is expressed in Year-0 dollars.

Step 1: The numerator is the present value of the growing annuity of annual outage costs (EAL_{out} growing at g_{outages}) over T_{lifetime} years, as of T_{COD} —the lump sum at T_{COD} equivalent to the stream from T_{COD} to end of life. Value at T_{COD} :

$$\text{PV at } T_{\text{COD}} = EAL_{\text{out}} \times \frac{1 - \left(\frac{1 + g_{\text{outages}}}{1 + r_{\text{social}}} \right)^{T_{\text{lifetime}}}}{r_{\text{social}} - g_{\text{outages}}}.$$

Step 2: The denominator discounts that lump sum from T_{COD} back to Year 0 by $(1 + r_{\text{social}})^{T_{\text{delay}} + T_{\text{construction}}}$.

The real (present value) cost is therefore:

$$C_{\text{outage}} = \frac{EAL_{\text{out}} \times \frac{1 - \left(\frac{1 + g_{\text{outages}}}{1 + r_{\text{social}}} \right)^{T_{\text{lifetime}}}}{r_{\text{social}} - g_{\text{outages}}}}{(1 + r_{\text{social}})^{T_{\text{delay}} + T_{\text{construction}}}}.$$

Note on VoLL. Value of lost load (VoLL) is the economic value placed on unserved energy (\$/MWh). The 10-tier piecewise schedule reflects the empirical pattern from customer surveys: per-MWh costs decline sharply in the first hours as initial disruption costs are amortized, reach a minimum around 8 hours, then rise as extended outages trigger cascading societal costs. The piecewise VoLL schedule applies to the load-shed fraction $\rho \cdot MW_{\text{lost}}$; the redispatch fraction $(1 - \rho) \cdot MW_{\text{lost}}$ is valued at the flat rate $c_{\text{redispatch}}$. Capacity at risk ϕ (e.g. 1 for radial, < 1 for meshed/redundant) is the fraction of line capacity assumed lost per outage event.

Default calibration

The default overhead outage model uses a decomposition $\lambda = 0.3 + 0.012 \times L$, giving an effective rate of $\sim 0.015/\text{mi}$ for a 100-mile line, $\sim 0.027/\text{mi}$ for a 20-mile line, and $\sim 0.013/\text{mi}$ for a 300-mile line. The terminal component (0.3 outages/yr) captures length-independent substation and protection equipment failures ($\sim 25\%$ of EHV outages; (Landgren 1986; Adler et al. 1994)); the per-mile component (0.012/mi/yr) reflects exposure-dependent conductor and insulator failures, informed by Canadian Electricity Association data (240 kV: 0.014/mi/yr; Koval 1996), Polish TSO multi-year averages (220 kV: 0.012/mi/yr; Pijarski and Belowski 2025), and U.S. national averages across voltage classes ($\sim 0.010/\text{mi/yr}$; NERC 2025). The underground repair duration multiplier $k_{\text{underground}} = 80$ (effective duration $6 \text{ h} \times 80 = 480 \text{ h} \approx 20$ days) reflects CIGRE Technical Brochure 815 (2020) average XLPE land-cable repair times, consistent with Ofgem/GHD (2020) modeled repair durations (20 days XLPE, 40 days mass-impregnated) and Wisconsin PSC guidance characterizing underground repairs as spanning weeks to months. The subsea outage rate default $r_{\text{subsea}} = 0.0007$ outages/(mi·yr) reflects Ofgem/GHD (2020) risk-assessed total failure rate data (0.000615 failures/(yr·cct-km), combined internal and external faults) and ENTSO-E/Europacable (2019) HVDC cable-only reliability data (0.07 faults/100 km/yr), both consistent with CIGRE Technical Brochure 815 (2020), which documents a reduction in submarine cable failure rates relative to the earlier TB 379 (2009) survey. The subsea duration multiplier $k_{\text{subsea}} = 250$ (effective duration ~ 62 days) reflects industry-consensus repair durations of 3–5 months for submarine export cable faults, documented in IEA Wind TEM95 proceedings (McKeever and Wyatt 2019) and CIGRE TB 815 (CIGRE Working Group B1.10 2020). The underground land cable rate ($\lambda_{\text{UG}} = 0.004/\text{mi/yr}$) draws on CIGRE Technical Brochure 379 (2009) and its update TB 815 (2020) (CIGRE Working Group B1.10 2020), which report XLPE land cable internal failure rates of 0.00069–0.00088/cct-km. The CTCC value (0.0025/km) is approximately 2–3 $\times$ the European cable-body baseline, reflecting a U.S. system-level rate that includes joint, termination, and third-party-damage failures—consistent with NERC TADS underground circuit outage data of ~ 0.003 – $0.005/\text{mi/yr}$ (2013–2018 average). Value of lost load tiers are directly calibrated to the ERCOT system-wide customer survey (Gibbons and Sergici 2024), which reports VoLL by duration from 1 to 16 hours (e.g. \$36,620/MWh at 1 hour, \$10,708/MWh at 8 hours, 2025\$). Beyond the 16-hour survey ceiling, VoLL is extrapolated using the three-term model $V(t) = 32,210/t + 3,339 + 492t$, capped at 30 days. The LBNL ICE Calculator (Sullivan, Schellenberg, and Blundell 2015; Larsen et al. 2025) provides corroborating estimates by customer class. The load-shed fraction ρ is determined by AC/DC classification ($\rho = 0.05$ for AC, $\rho = 0.80$ for DC). The redispatch cost $c_{\text{redispatch}}$ defaults to \$20/MWh (Millstein et al. 2022); the parameter may be varied for projects with known congestion exposure.

Risk cost total

The two risk-cost components above sum to the risk cost bucket. Risk costs are probabilistic and event-driven; they are not AFUDC-eligible and do not enter rate base.

$$C_{\text{risk}} = C_{\text{wf}} + C_{\text{outage}}.$$

Component PV equations: C_{wf} (§B.3.1), C_{outage} (§B.3.2).

Note on independence of wildfire and outage costs. A single wildfire event can simultaneously cause property and environmental damage (counted in C_{wf}) and a transmission outage (counted in C_{outage}), either directly through line damage or indirectly through preventive de-energization (PSPS). Despite this shared triggering event, independent treatment is defensible because the two cost categories capture distinct consequence dimensions: C_{wf} values property destruction, suppression expenditures, and environmental losses, whereas C_{outage} values the economic cost of unserved energy to consumers via VoLL. Additive treatment may modestly overstate total risk cost for projects in high-wildfire zones where fire-caused outages constitute a significant fraction of all outage events. However, for the majority of transmission projects outside extreme fire corridors, outage costs are driven primarily by weather, equipment failure, and vegetation contact unrelated to wildfire, and the overlap between the two cost streams is small.

B.4 Emissions Costs ($C_{\text{emissions}}$)

This subsection defines the emissions cost bucket: $C_{\text{emissions}}^P = C_{\text{comp,emissions}}^P$ (loss-compensation emissions only). The facilitated emissions costs $C_{\text{fac,emissions}}^P$ and $C_{\text{fac,emissions}}^{\text{no}}$ are intermediate quantities whose difference forms the avoided-emissions benefit $B_{\text{avoided,emissions}}^P$ (§C.3). Both calculations use the same rate-based mix-evolution equations, emission intensities $I_{j,k}$, societal costs $c_k(\tau)$, and social discount rate r_{social} . None of these costs are AFUDC-eligible or included in rate base. Note: emissions reductions are not among the seven required benefit categories in FERC Order No. 1920 (May 2024) but are permitted as optional additional benefits consistent with Order Nos. 890 and 1000.

B.4.1 Loss-compensation emissions

This subsection defines $C_{\text{comp,emissions}}$: the societal cost of emissions (CO_2 , SO_x , NO_x) from extra generation used to compensate for transmission energy losses. Energy losses (E_{loss}) are defined in §B.2.4; this subsection only adds externality valuation through the grid's time-varying fuel mix and pollutant prices. Under the single-trajectory grid model (§B.4.2), shares $p_j(\tau)$ here are the with-line grid shares $p_j^{\text{with}}(\tau)$: the regional grid mix evolving under the line's influence during operations, starting from the shared grid state at COD. The without-line path $p_j^{\text{no}}(\tau)$ used by the facilitated-emissions and displacement subsections does not enter loss-compensation calculations.

Variable	Description
E_{loss}	Total energy losses (annual) (MWh/year). From §B.2.4.
α	Compensation share (dimensionless [0,1]). Fraction of E_{loss} compensated by additional generation.
TEC	Total energy compensated (annual) (MWh/year). $TEC = \alpha \times E_{\text{loss}}$.
j	Energy source index (index). e.g. coal, gas, solar, wind, hydro, nuclear.
$p_j^{\text{with}}(\tau)$	With-line grid share of source j in operating year τ (dimensionless). Grid state at COD, $p_j(T_{\text{COD}})$ (§B.4.2), evolved at the post-COD per-source mix-evolution rates g_j^{post} ; $\sum_j p_j^{\text{with}}(\tau) = 1$. See §B.4.2 for the without-line path and the COD handoff.
$I_{j,k}$	Emission intensity (kg/MWh). Emissions of pollutant k per MWh from source j . Lifecycle medians from Bruckner et al. (2014), Table A.III.2.
k	Pollutant index (index). CO_2 , SO_x , NO_x .
$E_{k,\tau}$	Emissions of pollutant k in year τ (kg). From compensated energy and mix.
c_k	Societal cost per kg of pollutant k (\$/kg). Year-1 level, 2025\$. Default: \$236/t CO_2 (U.S. Environmental Protection Agency 2023b; Rennert et al. 2022); SO_x \$76/kg, NO_x \$143/kg (EPA BPT 2024 (U.S. Environmental Protection Agency 2023a), CPI-U escalated from 2020\$ to 2025\$, $\times 1.2438$).
$g_{c,k}$	Real annual escalation rate for c_k (decimal). Default 0.02 for CO_2 , 0 for SO_x/NO_x .
$c_k(\tau)$	Societal cost per kg in operating year τ (\$/kg). $c_k(1) \cdot (1 + g_{c,k})^{\tau-1}$.
$C_{k,\tau}$	Cost of pollutant k in year τ (\$). $E_{k,\tau} \times c_k(\tau)$.
C_τ	Total emissions cost in year τ (\$/year). Sum over pollutants.
T_{lifetime}	Project lifetime (years).
T_{COD}	Commercial operation date (year). First year of operation.
r_{social}	Social discount rate (decimal). Used to discount this externality (§A.2).
$C_{\text{comp,emissions,nominal}}$	Nominal lifetime loss-compensation emissions cost (\$). Undiscounted sum over lifetime.
$C_{\text{comp,emissions}}$	Loss-compensation emissions cost (PV) (\$). PV of annual stream from T_{COD} over T_{lifetime} .

Variables Default emission intensities $I_{j,k}$ (kg/MWh) by fuel type are lifecycle medians from IPCC AR5 (Bruckner et al. 2014), Table A.III.2:

Table 4: Default emission intensities $I_{j,k}$ (kg/MWh) by fuel type.

Fuel	CO ₂	SO _x	NO _x
Coal	820	13.0	4.0
Oil	650	10.0	3.5
Natural gas	450	0.1	0.2
Solar	48	0	0
Wind	12	0	0
Hydro	24	0	0
Nuclear	12	0	0

Equations

1. Total energy compensated (annual).

$$TEC = \alpha \times E_{\text{loss}}.$$

2. Energy mix by operating year. For each operating year $\tau = 1, \dots, T_{\text{lifetime}}$, with-line grid shares $p_j^{\text{with}}(\tau)$ evolve from the grid state at COD, $p_j(T_{\text{COD}})$ (§B.4.2, Eq. 2), at the post-COD rates g_j^{post} , normalized so

$$\sum_j p_j^{\text{with}}(\tau) = 1.$$

3. Emissions in operating year τ (by pollutant).

$$E_{k,\tau} = TEC \times \sum_j p_j^{\text{with}}(\tau) \times I_{j,k}.$$

4. Societal cost schedule.

$$c_k(\tau) = c_k(1) \cdot (1 + g_{c,k})^{\tau-1}.$$

5. Cost in operating year τ (by pollutant).

$$C_{k,\tau} = E_{k,\tau} \times c_k(\tau).$$

6. Total emissions cost in operating year τ .

$$C_\tau = \sum_k C_{k,\tau}.$$

7. Nominal (undiscounted) lifetime cost.

$$C_{\text{comp,emissions,nominal}} = \sum_{\tau=1}^{T_{\text{lifetime}}} C_\tau.$$

8. Present value.

$$C_{\text{comp,emissions}} = \sum_{\tau=1}^{T_{\text{lifetime}}} \frac{C_\tau}{(1 + r_{\text{social}})^{T_{\text{COD}} + \tau - 1}}.$$

Time-varying societal cost and discounting. The societal cost $c_k(\tau)$ escalates in real terms to reflect the increasing marginal damage of emissions as cumulative warming grows (Rennert et al. 2022). The escalation rate $g_{c,k}$ is distinct from the social discount rate r_{social} : $g_{c,k}$ reflects the physical trajectory of damage costs, while r_{social} discounts the annual cost stream to the project’s present value. These operate on different objects and do not constitute double-discounting.

Both C_{energy} and $C_{\text{comp,emissions}}$ originate from E_{loss} (physical transmission losses), but they use different valuations: C_{energy} applies the value of load to the MWh of lost energy; $C_{\text{comp,emissions}}$ applies externality prices to the pollutant emissions from the make-up generation. They measure different economic effects and do not double-count.

Grid mix assumption for loss compensation. Loss-compensation emissions are computed using the post-COD regional grid mix, reflecting marginal system dispatch. For radial lines delivering a specific resource (e.g., a dedicated wind corridor), this assumes losses are compensated by the regional fleet rather than by the resource itself. This is conservative (the regional fleet is more carbon-intensive than the dedicated resource) and consistent with the system-dispatch perspective: the generating resource produces independently of line losses, and the system compensates the delivery shortfall at the margin.

B.4.2 Facilitated emissions

The absolute facilitated emissions costs $C_{\text{fac,emissions}}^P$ and $C_{\text{fac,emissions}}^{\text{no}}$ are intermediate quantities computed here; they do not enter the cost bucket directly. Their difference forms the avoided-emissions benefit $B_{\text{avoided,emissions}}^P$ (§C.3).

This subsection defines $C_{\text{fac,emissions}}^{(s)}$: the absolute social cost of emissions from trajectory s 's grid path over the deliverable energy $E_{\text{delivered,annual}}$ (§C.2.1), under the single-trajectory grid model. There is one regional grid, starting from initial shares $p_j(0)$. Absent the line, it evolves continuously at the pre-COD rates g_j^{pre} ; with the line's influence, it evolves at the post-COD rates g_j^{post} . The grid's state at COD, $p_j(T_{\text{COD}})$, is the without-line trajectory evaluated at T_{COD} — the grid is whatever it evolved into during delay plus construction, since the line has not yet influenced it. Both operational trajectories start from that same COD state: $C_{\text{fac,emissions}}^P$ (with-line) evolves it forward at g_j^{post} ; $C_{\text{fac,emissions}}^{\text{no}}$ (without-line) evolves it forward at g_j^{pre} . Both enter the avoided-emissions calculation (§C.3).

Non-double-counting with loss compensation. The loss-compensation subsection (§B.4.1) and this subsection operate on disjoint MWh pools: E_{loss} (energy dissipated along the line) vs. $E_{\text{delivered,annual}}$ (throughput reaching the load end). Setting $\alpha = 0$ removes the loss-compensation stream without affecting facilitated emissions, and vice versa. No MWh appears in both formulas.

Non-double-counting with delivered-energy market benefit. The same $E_{\text{delivered,annual}}$ carries one market valuation (v_{load} in §C.2.1) and one externality valuation (c_k per kg here). These measure different economic objects: willingness-to-pay for delivered energy vs. social cost of pollutant emissions.

Transparency: value-of-load internalization. If v_{load} already partially reflects emissions costs (e.g. through carbon pricing, cap-and-trade allowance costs, or renewable energy credits), part of the externality may already be captured in the delivered-energy benefit. To avoid double-counting, either (a) define v_{load} net of carbon/emissions adders, or (b) define c_k to exclude externalities already reflected in the value of load. The default assumption is that v_{load} reflects a wholesale-level willingness to pay that does not fully internalize the societal externalities valued by c_k .

Variable	Description
$E_{\text{delivered,annual}}$	Deliverable energy (MWh/year). From §C.2.1. Constant over lifetime.
j	Energy source index (index). Same as §B.4.1.
k	Pollutant index (index). CO_2 , SO_x , NO_x ; same as §B.4.1.
$p_j(0)$	Initial grid share of source j (dimensionless). Exogenous initial regional generation shares; $\sum_j p_j(0) = 1$. Single regional grid, shared by both trajectories.
g_j^{pre}	Per-source annual rate of change without the line (exogenous). Governs the without-line trajectory continuously from year 0.
g_j^{post}	Per-source annual rate of change with the line (exogenous). Governs the with-line trajectory from COD onward (and hypothetically during delay; §C.3.1).
$p_j(T_{\text{COD}})$	Grid share of source j at COD (dimensionless). $= p_j^{\text{no}}(T_{\text{COD}})$: the without-line trajectory evaluated at COD. Shared starting point for both operational instances.
$p_j^{\text{with}}(\tau)$	With-line grid share of source j , operating year τ . $p_j(T_{\text{COD}})$ evolved at g_j^{post} ; same path used by loss-comp (§B.4.1).
$p_j^{\text{no}}(\tau)$	Without-line grid share of source j , operating year τ . $p_j(T_{\text{COD}})$ evolved at g_j^{pre} ; used here and by displacement (§C.3).
$I_{j,k}$	Emission intensity (kg/MWh). Shared with loss-compensation emissions (§B.4.1); lifecycle medians as cited there (e.g. Bruckner et al.).
$EF_k^{(s)}(\tau)$	Implied emission factor (kg/MWh). For instance $s \in \{\text{with}, \text{no}\}$.
$E_{k,\tau}^{(s)}$	Emissions mass under instance s , year τ (kg).
$C_{k,\tau}^{(s)}$	Emissions cost under instance s , year τ (\$).
$PV_k^{(s)}$	PV emissions cost for pollutant k , instance s (\$).
c_k	Societal cost per kg of pollutant k (\$/kg). Year-1 level; same as §B.4.1.
$g_{c,k}$	Real annual escalation rate for c_k (decimal). Same as §B.4.1. Default 0.02 for CO_2 , 0 for SO_x/NO_x .
$c_k(\tau)$	Societal cost per kg in operating year τ (\$/kg). $c_k(1) \cdot (1 + g_{c,k})^{\tau-1}$.
r_{social}	Social discount rate (decimal). Same as §B.4.1.
T_{COD}	Commercial operation date (year). First year of operation.
T_{lifetime}	Project lifetime (years).
$C_{\text{fac,emissions}}^{(s)}$	Absolute facilitated emissions cost, instance s (\$). $= \sum_k PV_k^{(s)}$. Intermediate; does not enter $C_{\text{emissions}}^P$ directly.

Variables

Equations

1. Scope (deliverable energy). The scoped MWh is the deliverable energy from §C.2.1, constant over the project lifetime:

$$H_{\mathcal{S},\tau} = E_{\text{delivered,annual}}.$$

2. Grid state at COD (single-trajectory handoff). Absent the line, the grid evolves continuously at g_j^{pre} from the initial shares $p_j(0)$. This same without-line path also defines the grid's actual state the moment the line reaches commercial operation. Step 1—compound growth/decay from year 0 to T_{COD} :

$$\tilde{p}_j(T_{\text{COD}}) = p_j(0) \cdot (1 + g_j^{\text{pre}})^{T_{\text{COD}}-1}.$$

Step 2—renormalize:

$$p_j(T_{\text{COD}}) = \frac{\tilde{p}_j(T_{\text{COD}})}{\sum_{j'} \tilde{p}_{j'}(T_{\text{COD}})}.$$

$p_j(T_{\text{COD}})$ is the shared starting point for *both* operational trajectories below — the with-line and without-line mixes are identical up through COD and diverge only afterward.

3. Operational mix evolution (both instances, from the COD state). For each trajectory $s \in \{\text{with}, \text{no}\}$, starting from the common COD state $p_j(T_{\text{COD}})$, the same rate-based compound-growth equations used in §B.4.1 yield operating-year time paths. Step 1—compound growth/decay:

$$\tilde{p}_j^{(s)}(\tau) = p_j(T_{\text{COD}}) \cdot (1 + g_j^{(s)})^{\tau-1}, \quad \tau = 1, \dots, T_{\text{lifetime}}.$$

Step 2—renormalize each year:

$$p_j^{(s)}(\tau) = \frac{\tilde{p}_j^{(s)}(\tau)}{\sum_{j'} \tilde{p}_{j'}^{(s)}(\tau)}.$$

The with-line trajectory uses $g_j^{(\text{with})} = g_j^{\text{post}}$; the without-line trajectory uses $g_j^{(\text{no})} = g_j^{\text{pre}}$.

4. Implied emission factor. For $s \in \{\text{with, no}\}$ and pollutant k :

$$EF_k^{(s)}(\tau) = \sum_j p_j^{(s)}(\tau) I_{j,k}.$$

5. Absolute emissions mass per year (by instance). For instance s and pollutant k :

$$E_{k,\tau}^{(s)} = E_{\text{delivered,annual}} \times EF_k^{(s)}(\tau).$$

6. Societal cost schedule.

$$c_k(\tau) = c_k(1) \cdot (1 + g_{c,k})^{\tau-1}.$$

7. Absolute emissions cost per year.

$$C_{k,\tau}^{(s)} = E_{k,\tau}^{(s)} \times c_k(\tau).$$

8. Present value per pollutant (by instance). Discounted at r_{social} with the same index convention as §B.4.1:

$$PV_k^{(s)} = \sum_{\tau=1}^{T_{\text{lifetime}}} \frac{C_{k,\tau}^{(s)}}{(1 + r_{\text{social}})^{T_{\text{COD}} + \tau - 1}}.$$

9. Total across pollutants — absolute facilitated emissions cost.

$$C_{\text{fac,emissions}}^{(s)} = \sum_k PV_k^{(s)}.$$

Pollutants: CO_2 , SO_x , NO_x —the same set as §B.4.1. $C_{\text{fac,emissions}}^P$ (with-line, $s = \text{with}$) and $C_{\text{fac,emissions}}^{\text{no}}$ are intermediate quantities; their difference forms $B_{\text{avoided,emissions}}^P$ (§C.3).

Emissions cost aggregate

$$C_{\text{emissions}}^P = C_{\text{comp,emissions}}^P.$$

See §B.4.1 for $C_{\text{comp,emissions}}^P$. The facilitated emissions costs $C_{\text{fac,emissions}}^P$ and $C_{\text{fac,emissions}}^{\text{no}}$ (§B.4.2) are intermediate quantities; their difference enters the benefit total as $B_{\text{avoided,emissions}}^P$ (§C.3).

C Benefits

This section defines the three benefit buckets. Remedial benefits (B_{remedial}) are relief of pre-existing system harms: curtailment reduction (§C.1.1) and congestion reduction (§C.1.2). Enabling benefits (B_{enabling}) are new productive value the line creates: the benefit of delivered energy (§C.2.1). Avoided-emissions benefits ($B_{\text{avoided,emissions}}$) capture the clean-generation displacement incentive: the reduction in societal emissions costs when the project's facilitated generation mix is cleaner than the no-line counterfactual (§C.3). Effective capacity relief and the Approach B congestion/curtailment split ($B_{\text{congestion,annual}}$, $B_{\text{curtailment,annual}}$) are defined in §A.4; this section only adds lifetime aggregation (nominal and real PV). Congestion and curtailment draw on the same single constraint, split by the congestion fraction f (§A.4). Benefits are not part of rate base or AFUDC.

C.1 Remedial Benefits (B_{remedial})

Remedial benefits are relief of pre-existing system harms: curtailment reduction and congestion reduction. $B_{\text{remedial}}^P = B_{\text{curtailment,lifetime}}^P + B_{\text{congestion,lifetime}}^P$.

C.1.1 Curtailment reduction benefit

This subsection defines the curtailment reduction benefit, which contributes to B_{remedial} (§D). The annual curtailment benefit $B_{\text{curtailment,annual}}$ is defined in §A.4 (Approach B); this subsection only adds lifetime aggregation (nominal and real PV). Curtailment is when generators (e.g. renewables) must cut output because the grid cannot deliver it. The curtailment reduction benefit is the value of the fraction $(1 - f)$ of constrained hours that resolve as curtailment, valued at $\gamma_{\text{curtailment}}$ (\$/MWh). Annual benefit is summed over the project lifetime in nominal terms or discounted to present value at real WACC from the commercial operation date.

Variable	Description
$B_{\text{curtailment,annual}}$	Annual benefit from curtailment reduction (\$/year). From §A.4.
T_{lifetime}	Project lifetime (years).
T_{COD}	Commercial operation date (year index) (years). First year of operation; benefit stream starts here.
$r_{\text{WACC,real}}$	Real WACC (decimal). Used to discount benefits (§A.2).
g_{benefit}	Benefit price escalation rate (decimal). From §A.2.
$B_{\text{curtailment,lifetime,nominal}}$	Nominal lifetime curtailment benefit (\$).
$B_{\text{curtailment,lifetime}}$	Real (PV) lifetime curtailment benefit (\$).

Variables

Equations As in §A.4, $B_{\text{curtailment,annual}} = H \times (1 - f) \times \min(\Delta C_{\text{effective}}, X) \times \gamma_{\text{curtailment}}$.

1. Nominal lifetime curtailment benefit. Growing annual benefit at rate g_{benefit} over T_{lifetime} years:

$$B_{\text{curtailment,lifetime,nominal}} = B_{\text{curtailment,annual}} \times \frac{(1 + g_{\text{benefit}})^{T_{\text{lifetime}}} - 1}{g_{\text{benefit}}}.$$

At $g_{\text{benefit}} = 0$ this reduces to $B_{\text{curtailment,annual}} \times T_{\text{lifetime}}$.

2. Real (present value) lifetime curtailment benefit. Growing annual benefit from T_{COD} for T_{lifetime} years, discounted at $r_{\text{WACC,real}}$:

$$B_{\text{curtailment,lifetime}} = \frac{B_{\text{curtailment,annual}} \times \frac{1 - \left(\frac{1 + g_{\text{benefit}}}{1 + r_{\text{WACC,real}}} \right)^{T_{\text{lifetime}}}}{r_{\text{WACC,real}} - g_{\text{benefit}}}}{(1 + r_{\text{WACC,real}})^{T_{\text{delay}} + T_{\text{construction}}}}.$$

At $g_{\text{benefit}} = 0$ this reduces to the flat-annuity formula. The structure matches the wildfire/outage growing-annuity PV in §B.3.1.

Link: Both benefit streams (§C.1.2) draw on the same constrained hours H , exceedance X , and effective relief $\Delta C_{\text{effective}}$, split by the congestion fraction f (§A.4); this ensures the two benefits are consistent and not double-counted.

C.1.2 Congestion reduction benefit

This subsection defines the congestion reduction benefit, which contributes to B_{remedial} (§D). The annual congestion benefit $B_{\text{congestion,annual}}$ is defined in §A.4 (Approach B); this subsection only adds lifetime aggregation (nominal and real PV). Congestion is when the grid is short of capacity and cannot deliver all the power that is wanted, so value is lost (e.g. higher prices, redispatch). The congestion reduction benefit is the value of the fraction f of constrained hours that resolve as redispatch, valued at $\gamma_{\text{congestion}}$ (\$/MWh). Annual benefit is summed over the project lifetime in nominal terms or discounted to present value at real WACC from the commercial operation date.

Variable	Description
$B_{\text{congestion,annual}}$	Annual benefit from congestion reduction (\$/year). From §A.4.
T_{lifetime}	Project lifetime (years).
T_{COD}	Commercial operation date (year index) (years). First year of operation; benefit stream starts here.
$r_{\text{WACC,real}}$	Real WACC (decimal). Used to discount benefits (§A.2).
g_{benefit}	Benefit price escalation rate (decimal). From §A.2.
$B_{\text{congestion,lifetime,nominal}}$	Nominal lifetime congestion benefit (\$).
$B_{\text{congestion,lifetime}}$	Real (PV) lifetime congestion benefit (\$).

Variables

Equations As in §A.4, $B_{\text{congestion,annual}} = H \times f \times \min(\Delta C_{\text{effective}}, X) \times \gamma_{\text{congestion}}$.

1. Nominal lifetime congestion benefit. Growing annual benefit at rate g_{benefit} over T_{lifetime} years:

$$B_{\text{congestion,lifetime,nominal}} = B_{\text{congestion,annual}} \times \frac{(1 + g_{\text{benefit}})^{T_{\text{lifetime}}} - 1}{g_{\text{benefit}}}.$$

At $g_{\text{benefit}} = 0$ this reduces to $B_{\text{congestion,annual}} \times T_{\text{lifetime}}$.

2. Real (present value) lifetime congestion benefit. Growing annual benefit from T_{COD} for T_{lifetime} years, discounted at $r_{\text{WACC,real}}$:

$$B_{\text{congestion,lifetime}} = \frac{B_{\text{congestion,annual}} \times \frac{1 - \left(\frac{1 + g_{\text{benefit}}}{1 + r_{\text{WACC,real}}} \right)^{T_{\text{lifetime}}}}{r_{\text{WACC,real}} - g_{\text{benefit}}}}{(1 + r_{\text{WACC,real}})^{T_{\text{delay}} + T_{\text{construction}}}}.$$

At $g_{\text{benefit}} = 0$ this reduces to the flat-annuity formula.

Link: Effective relief $\Delta C_{\text{effective}}$ and the congestion-fraction split f are defined in §A.4 and used identically in §C.1.1. The same single-constraint decomposition ensures congestion and curtailment benefits are consistent and not double-counted.

C.2 Enabling Benefits (B_{enabling})

Enabling benefits are new productive value the line creates: the benefit of delivered energy. $B_{\text{enabling}}^P = B_{\text{delivered,lifetime}}^P$.

C.2.1 Benefit of delivered energy

This subsection defines the benefit of delivered energy, which constitutes B_{enabling} (§D): the societal value of the energy the project enables to be delivered each year. For greenfield, deliverable capacity equals effective capacity as in §A.4, $\Delta C_{\text{effective}} = \phi \times C_{\text{new}}$. For reconductoring, deliverable capacity is the additional capacity the upgrade enables, $C_{\text{new}} - C_{\text{old}}$ (same as $\Delta C_{\text{effective}}$ in §A.4 for reconductoring); this captures the value of additional energy the upgrade enables, including any headroom beyond clearing the constraint. Deliverable energy per year is deliverable capacity $\times$ line utilization $\times$ hours per year (MWh/year), valued at value of load v_{load} (\$/MWh). Level annual benefit from COD, discounted at real WACC. This benefit is not part of rate base or AFUDC and is not double-counted with congestion or curtailment (those value relief of constraints; this values throughput).

Variable	Description
$\Delta C_{\text{effective}}$	Effective capacity relief (MW). From §A.4. Greenfield: ϕC_{new} ; reconductoring: $C_{\text{new}} - C_{\text{old}}$.
u	Line utilization (dimensionless). From §B.2.4. $0 \leq u \leq 1$.
H	Hours per year (h/year). e.g. 8760. From §B.2.4.
v_{load}	Value of load (\$/MWh). Marginal value of energy to end-use load. From §B.2.5.
$E_{\text{delivered,annual}}$	Deliverable energy per year (MWh/year).
$B_{\text{delivered,annual}}$	Annual benefit from delivered energy (\$/year).
T_{lifetime}	Project lifetime (years).
T_{COD}	Commercial operation date (year index) (years). First year of operation; benefit stream starts here.
$r_{\text{WACC,real}}$	Real WACC (decimal). Used to discount benefits (§A.2).
g_{benefit}	Benefit price escalation rate (decimal). From §A.2.
$B_{\text{delivered,lifetime,nominal}}$	Nominal lifetime benefit from delivered energy (\$).
$B_{\text{delivered,lifetime}}$	Real (PV) lifetime benefit from delivered energy (\$).

Variables

Equations

1. Deliverable energy (annual).

$$E_{\text{delivered,annual}} = \Delta C_{\text{effective}} \times u \times H.$$

2. Annual benefit.

$$B_{\text{delivered,annual}} = E_{\text{delivered,annual}} \times v_{\text{load}}.$$

3. Nominal lifetime benefit.

Growing annual benefit at rate g_{benefit} over T_{lifetime} years:

$$B_{\text{delivered,lifetime,nominal}} = B_{\text{delivered,annual}} \times \frac{(1 + g_{\text{benefit}})^{T_{\text{lifetime}}} - 1}{g_{\text{benefit}}}.$$

At $g_{\text{benefit}} = 0$ this reduces to $B_{\text{delivered,annual}} \times T_{\text{lifetime}}$.

4. Real (present value) lifetime benefit. Growing annual benefit from T_{COD} for T_{lifetime} years, discounted at $r_{\text{WACC,real}}$:

$$B_{\text{delivered,lifetime}} = \frac{B_{\text{delivered,annual}} \times \frac{1 - \left(\frac{1 + g_{\text{benefit}}}{1 + r_{\text{WACC,real}}} \right)^{T_{\text{lifetime}}}}{r_{\text{WACC,real}} - g_{\text{benefit}}}}{(1 + r_{\text{WACC,real}})^{T_{\text{delay}} + T_{\text{construction}}}}.$$

At $g_{\text{benefit}} = 0$ this reduces to the flat-annuity formula.

Default calibration

$g_{\text{benefit}} = 0.01$ (1% real/yr). This reflects an older CAISO Transmission Economic Assessment Methodology (TEAM) practice of applying 1% real escalation to benefits extrapolated beyond the production-cost simulation horizon (Chang, Pfeifenberger, and Hagerty 2013). CAISO’s current (2024–2025 Transmission Plan) methodology uses $g_{\text{benefit}} = 0$ for post-horizon benefit extension (California Independent System Operator 2025). Historical real U.S. electricity prices were flat to declining 2010–2020 (Lam 2026), but the CTCC applies escalation uniformly across the entire 30–50 year project lifetime (no simulated-year base), making a conservative positive default appropriate. Sensitivity: 0–4% (Figure F9).

Link: $\Delta C_{\text{effective}}$ is defined in §A.4; v_{load} is defined in §B.2.5. Total societal benefits comprise B_{remedial} (congestion + curtailment), B_{enabling} (delivered energy), and $B_{\text{avoided,emissions}}$ (displacement); see §D for bucket definitions.

C.3 Displacement avoided emissions

This subsection derives the avoided-emissions benefit $B_{\text{avoided,emissions}}^P$: the difference between the no-line and with-line absolute facilitated emissions costs from §B.4.2. Under the single-trajectory grid model, both trajectories start from the identical grid state at COD and diverge only afterward, as the line’s influence (g_j^{post}) pulls the grid onto a different path than it would have followed on its own (g_j^{pre}). Transmission projects that facilitate greater enablement of clean generation leading to displacement of fossil fuels produce a measurable reduction in the social cost of emissions; $B_{\text{avoided,emissions}}^P$ is the formal quantification of that displacement in social-cost terms. A positive $B_{\text{avoided,emissions}}^P$ means the with-line grid trajectory is cleaner than the without-line counterfactual—an avoided externality cost that enters the benefit bucket $B_{\text{avoided,emissions}}$. Not part of rate base or AFUDC.

Variables

Variable	Description
$C_{\text{fac,emissions}}^P$	Facilitated emissions cost, project (\$). From §B.4.2.
$C_{\text{fac,emissions}}^{\text{no}}$	Facilitated emissions cost, no-line (\$). From §B.4.2.
$E_{k,\tau}^{(s)}$	Emissions mass under instance s , year τ (kg). From §B.4.2.
$E_{k,\tau}^{\text{avoid}}$	Avoided emissions mass in year τ (kg). $= E_{k,\tau}^{\text{no}} - E_{k,\tau}^{\text{with}}$.
c_k	Societal cost per kg of pollutant k (\$/kg). Same as §B.4.1.
$C_{k,\tau}^{\text{avoid}}$	Avoided emissions cost in year τ (\$).
r_{social}	Social discount rate (decimal). Same as §B.4.1.
T_{COD}	Commercial operation date (year).
T_{lifetime}	Project lifetime (years).
PV_{avoid}^k	PV avoided cost for pollutant k (\$).
$B_{\text{avoided,emissions}}^P$	Avoided-emissions benefit (\$). $= C_{\text{fac,emissions}}^{\text{no}} - C_{\text{fac,emissions}}^P$. Enters benefit bucket $B_{\text{avoided,emissions}}$.

Equations

1. Avoided emissions mass per year. For pollutant k , using the per-instance emissions mass from §B.4.2:

$$E_{k,\tau}^{\text{avoid}} = E_{k,\tau}^{\text{no}} - E_{k,\tau}^{\text{with}} = E_{\text{delivered,annual}} \times (EF_k^{\text{no}}(\tau) - EF_k^{\text{with}}(\tau)).$$

Positive $E_{k,\tau}^{\text{avoid}}$ means fewer emissions under the with-line grid trajectory than under the without-line counterfactual. Negative values (with-line trajectory dirtier) are allowed.

2. Avoided emissions cost per year.

$$C_{k,\tau}^{\text{avoid}} = E_{k,\tau}^{\text{avoid}} \times c_k(\tau).$$

3. PV of avoided cost per pollutant.

$$PV_k^{\text{avoid}} = PV_k^{\text{no}} - PV_k^{\text{with}} = \sum_{\tau=1}^{T_{\text{lifetime}}} \frac{C_{k,\tau}^{\text{avoid}}}{(1 + r_{\text{social}})^{T_{\text{COD}} + \tau - 1}}.$$

4. Total avoided-emissions benefit.

$$B_{\text{avoided,emissions}}^P = C_{\text{fac,emissions}}^{\text{no}} - C_{\text{fac,emissions}}^P = \sum_k PV_k^{\text{avoid}}.$$

Positive = cleaner with-line trajectory = avoided externality. $B_{\text{avoided,emissions}}^P$ enters $B_{\text{avoided,emissions}}$ (§D).

Pairwise corollary. $\Delta B_{\text{avoided,emissions}}^{P:B} = B_{\text{avoided,emissions}}^P - B_{\text{avoided,emissions}}^B = C_{\text{fac,emissions}}^B - C_{\text{fac,emissions}}^P$ (the no-line reference term cancels). See §D for incremental reporting.

C.3.1 Displacement Delay Emissions Cost

Each year of project delay represents a year of foregone emissions displacement. Under the single-trajectory grid model (§B.4.2), both trajectories start from the same initial grid shares $p_j(0)$ at year 0: the without-line path evolves at g_j^{pre} , continuing the trajectory that is later used as the grid's actual state at COD; the with-line path evolves *hypothetically* at g_j^{post} , as if the line's influence on the grid had begun at year 0 — "what if the line had been built on time?" The displacement delay cost values the foregone benefit of that hypothetical, computing for each delay year the emissions reduction that would have occurred had the project been operational since year 0:

$$C_{\text{displacement,delay}} = \sum_{t=1}^{T_{\text{delay}}} \frac{B_{\text{displacement}}(t)}{(1 + r_{\text{social}})^t},$$

where

$$B_{\text{displacement}}(t) = E_{\text{delivered}} \times \sum_k \left[\sum_j (p_j^{\text{no}}(t) - p_j^{\text{with}}(t)) \times I_{j,k} \right] \times c_k(t),$$

$$p_j^{\text{no}}(t) \propto p_j(0)(1 + g_j^{\text{pre}})^{t-1}, \quad p_j^{\text{with}}(t) \propto p_j(0)(1 + g_j^{\text{post}})^{t-1} \quad (\text{both renormalized each year}).$$

$c_k(t) = c_k(0) \times (1 + g_k)^t$ reflects growing social costs of carbon (Rennert et al. 2022; Office of Management and Budget 2023).

Single-trajectory handoff at COD. At COD, the hypothetical with-line path used above is discarded and replaced by the grid's actual state, $p_j(T_{\text{COD}}) = p_j^{\text{no}}(T_{\text{COD}})$ — the grid IS whatever it evolved into without the line during delay plus construction, since the line has not yet influenced it (§B.4.2, Eq. 2). The operational-period displacement benefit (§C.3) then evolves both the with-line and without-line paths forward from that identical COD state, rather than restarting from year-0 percentages. Consequently, a longer delay leaves the grid further along its without-line decarbonization path by the time the project reaches COD, which mechanically reduces the operational displacement benefit available to a delayed project relative to one that reached COD sooner — consistent with the intuition that a cleaner starting grid has less dirty generation left to displace.

D Reporting Framework

This section assembles the component present values defined in the four cost bucket subsections (§B.1–§B.4) into two reporting scalars (scenario-internal net benefit and incremental net benefit vs a stated baseline) and defines the sign conventions used throughout the paper.

Bucket totals and grand cost

Cost components are organized into four reporting buckets (defined in §B.1.3, §B.2.8, §B.3.2, and §B.4):

Bucket	Equation	Constituent components
C_{hard}	$C_{\text{build}} + C_{\text{ROW,capital}} + C_{\text{env.mit}}$	§B.1.1, §B.1.2, §B.1.3
C_{soft}	$C_{\text{operational}} + C_{\text{energy}} + C_{\text{delay}}$	§B.2.1–§B.2.3, §B.2.4–§B.2.5, §B.2.6–§B.2.8
C_{risk}	$C_{\text{wf}} + C_{\text{outage}}$	§B.3
$C_{\text{emissions}}$	$C_{\text{comp,emissions}}$	§B.4.1

Grand total cost for scenario P :

$$C^P = C_{\text{hard}}^P + C_{\text{soft}}^P + C_{\text{risk}}^P + C_{\text{emissions}}^P.$$

$C_{\text{emissions}} (= C_{\text{comp,emissions}})$ includes loss-compensation emissions only (§B.4.1). The facilitated emissions costs $C_{\text{fac,emissions}}^P$ and $C_{\text{fac,emissions}}^{\text{no}}$ are intermediate quantities (§B.4.2); their difference enters the benefit total as $B_{\text{avoided,emissions}}^P$ (§C.3).

Benefit buckets and total benefit

The benefit streams from §C are organized into three reporting buckets:

Bucket	Equation	Constituent components
B_{remedial}	$B_{\text{curtailment,lifetime}} + B_{\text{congestion,lifetime}}$	§C.1.1, §C.1.2
B_{enabling}	$B_{\text{delivered,lifetime}}$	§C.2.1
$B_{\text{avoided,emissions}}$	$C_{\text{fac,emissions}}^{\text{no}} - C_{\text{fac,emissions}}^P$	§C.3

Total benefit for scenario P :

$$B^P = B_{\text{remedial}}^P + B_{\text{enabling}}^P + B_{\text{avoided,emissions}}^P.$$

B_{remedial} captures relief of pre-existing system harms (binding congestion, curtailed generation). B_{enabling} captures new productive value (delivered energy throughput). $B_{\text{avoided,emissions}}$ captures the clean-generation incentive: the reduction in societal emissions costs when the project's facilitated mix displaces a dirtier counterfactual (§C.3).

Scenario-internal net benefit

The scenario-internal net benefit answers “how does scenario P look on its own ledger?”:

$$\text{NB}^P = B^P - C^P.$$

Incremental net benefit vs a stated baseline

For pairwise comparisons, let B denote a stated baseline scenario. The incremental cost convention follows the avoided-cost sign rule: positive means scenario P has lower PV cost than baseline B .

Incremental cost (avoided-cost convention).

$$\Delta C^{P:B} = C^B - C^P \quad (\Delta C^{P:B} > 0 \Rightarrow P \text{ avoids cost relative to } B).$$

Componentwise:

$$\Delta C_{\text{hard}}^{P:B} = C_{\text{hard}}^B - C_{\text{hard}}^P, \quad \Delta C_{\text{soft}}^{P:B} = C_{\text{soft}}^B - C_{\text{soft}}^P, \quad \Delta C_{\text{risk}}^{P:B} = C_{\text{risk}}^B - C_{\text{risk}}^P, \quad \Delta C_{\text{emissions}}^{P:B} = C_{\text{emissions}}^B - C_{\text{emissions}}^P.$$

Incremental benefits.

$$\Delta B_{\text{remedial}}^{P:B} = B_{\text{remedial}}^P - B_{\text{remedial}}^B,$$

$$\Delta B_{\text{enabling}}^{P:B} = B_{\text{enabling}}^P - B_{\text{enabling}}^B,$$

$$\Delta B_{\text{avoided,emissions}}^{P:B} = B_{\text{avoided,emissions}}^P - B_{\text{avoided,emissions}}^B.$$

A positive ΔB means P delivers more benefit than baseline B in that channel.

Incremental net benefit.

$$\Delta \text{NB}^{P:B} = \Delta B^{P:B} + \Delta C^{P:B}.$$

Consistency check: $\Delta \text{NB}^{P:B} = \text{NB}^P - \text{NB}^B$, since $(B^P - B^B) - (C^P - C^B) = \Delta B^{P:B} + (C^B - C^P)$.

Risk and avoided risk. A lower risk cost under P than under B appears as a positive $\Delta C_{\text{risk}}^{P:B}$ in the incremental table. It is not added to B_{remedial} ; the incremental cost row is the avoided-risk story.

Emissions and displacement. The emissions bucket holds only loss-compensation emissions, $C_{\text{comp,emissions}}^P$. Displacement enters benefits, not costs.

The clean-generation incentive shows up in single-scenario NB via $B_{\text{avoided,emissions}}^P$, and in pairwise comparison via $\Delta B_{\text{avoided,emissions}}^{P:B}$ (the no-line term cancels).

A project with lower line losses generates a lower loss-compensation emissions cost. This contributes positively to $\Delta C_{\text{emissions}}^{P:B}$ when P is cheaper than baseline B .

Benefit-cost ratios

A benefit-cost ratio under perspective π is:

$$\text{BCR}_\pi = \frac{B_\pi^P}{C_\pi^P}.$$

$\text{BCR}_\pi > 1$ means benefits exceed costs under that perspective. The CTCC reports six core perspectives and three exclusion variants. Each answers a different question for a different audience; the choice of which BCR to prioritize depends on the stakeholder’s position relative to the project.

Perspective	Label	Numerator (B_π)	Denominator (C_π)	Question answered
<i>Societal</i> Societal	BCR_{soc}	B^P	C^P	Is this project a net positive for society?
<i>System (grid-operational)</i> Congestion relief	BCR_{cong}	B_{remedial}^P	$C_{\text{hard}}^P + C_{\text{operational}}^P + C_{\text{loss}}^P$	Does congestion/curtailment relief alone cover project costs?
<i>Firm</i> Utility / TSP	BCR_{util}	ATRR_{PV}	$\text{ATRR}_{\text{PV}} + C_{\text{base delay}}$	Does revenue cover all utility costs including delay?
Ratepayer	BCR_{rp}	$B_{\text{remedial}}^P + B_{\text{enabling}}^P$	$\text{ATRR}_{\text{PV}} + C_{\text{loss}}$	Do bill-relevant benefits exceed what ratepayers pay?
<i>Capital screening</i> Capital-only	BCR_{cap}	B^P	C_{hard}	How do benefits compare to capital investment alone?
Capital+delay	$\text{BCR}_{\text{cap+del}}$	B^P	$C_{\text{hard}} + C_{\text{delay}}$	What is the total pre-COD and capital exposure?
<i>Societal exclusion variants</i> Excl. avoided emissions	$\text{BCR}_{\text{excl av.em.}}$	$B^P - B_{\text{avoided,emissions}}^P$	C^P	What is the BCR under the FERC-minimum benefit set?
Excl. wildfire	$\text{BCR}_{\text{excl wf}}$	B^P	$C^P - C_{\text{wf}}$	What is the BCR for low-wildfire regions?
Excl. wildfire + outage	$\text{BCR}_{\text{excl wf+out}}$	B^P	$C^P - C_{\text{wf}} - C_{\text{outage}}$	What is the deterministic BCR (no risk costs)?

The following paragraphs justify each perspective, cite the regulatory and academic authority for its inclusion, and explain when $\text{BCR} < 1$ does and does not signal project failure.

Societal (BCR_{soc}). The societal BCR aggregates all quantifiable benefits against all costs with revenue excluded as a zero-sum transfer. This is the broadest reading of FERC Order No. 1920’s requirement that benefits be “evaluated comprehensively over a minimum 20-year horizon” (FERC 2024). Pfeifenberger recommends that planners “account for the full range of transmission projects’ benefits and use multi-value planning to comprehensively identify investments that cost-effectively address all categories of needs and benefits” (Pfeifenberger 2022). Revenue is excluded from the societal BCR because it is a transfer payment between utility and ratepayers—zero-sum from society’s perspective (Pfeifenberger et al. 2021). $\text{BCR}_{\text{soc}} < 1$ is a genuine negative signal: absent unquantified benefits (reliability, resource adequacy, option value), the project destroys net societal value.

Numerator discipline. Societal BCRs use B^P in the numerator. Gains from lower risk, emissions, capital, or delay show up as lower C_π in the denominator—not as extra benefit lines in the numerator. This is the same principle as the incremental avoided-cost convention: a positive $\Delta C^{P:B}$ means P avoids cost, not that P earns additional benefit. The full revenue requirement (ATRR_{PV} , §E) is a transfer; it enters perspective-specific views but not the societal NB.

Congestion relief (BCR_{cong}). This BCR uses only remedial benefits—congestion and curtailment relief—in the numerator, isolating the directly measurable grid-operational improvements. It corresponds to the screening metric used by ISO/RTO market-efficiency evaluations: PJM evaluates Market Efficiency Projects using a $BCR \geq 1.25$ threshold where benefits equal adjusted production cost (APC) savings from reduced congestion (Pfeifenberger et al. 2021); MISO applies the same threshold for its Market Efficiency Projects with APC savings as the primary benefit (MISO 2024a). The CTCC’s congestion-relief BCR is narrower than these ISO metrics because it excludes the value of enabling lower-cost dispatch (captured in the Ratepayer BCR). $BCR_{\text{cong}} < 1$ is expected for many project types—reliability projects, public policy projects, and projects justified by broader benefits will routinely fall below this threshold.

Utility / TSP (BCR_{util}). The Utility BCR is a pure financial viability test from the utility’s perspective: does the revenue the utility collects cover everything the utility pays? Both numerator and denominator use the utility’s own cost basis—no societal PV. The numerator is the full annual transmission revenue requirement ($ATRR_{\text{PV}}$: capital recovery + O&M + insurance + ROW rent), following the FERC cost-of-service formula (FERC 2026). The denominator is $ATRR_{\text{PV}} + C_{\text{base delay}}$: the same recoverable costs, plus pre-construction delay expenditures (legal, admin, permitting) that are not in rate base. Since the ATRR terms cancel:

$$BCR_{\text{util}} = \frac{ATRR_{\text{PV}}}{ATRR_{\text{PV}} + C_{\text{base delay}}} = 1.0 \text{ when } C_{\text{base delay}} = 0.$$

$BCR_{\text{util}} = 1.0$ is a structural identity for zero-delay regulated projects, not an approximation. Deviation below 1.0 measures exactly the fraction of the utility’s costs that the revenue requirement does not recover—the pre-COD delay penalty (FERC 1995).

Utility denominator scope. The utility denominator includes only costs the utility actually bears: the revenue requirement (ATRR, which covers capital recovery, O&M, insurance, and ROW rent) plus base delay costs ($C_{\text{base delay}}$: legal, admin, labor, regulatory, project management during the pre-construction period). Congestion and curtailment delay costs—the societal opportunity cost of constraints persisting during the delay period—are excluded because the utility does not bear them; they flow to market participants through LMP.

Ratepayer (BCR_{rp}). The Ratepayer BCR measures whether the benefits that flow to customer bills—reduced energy procurement costs from congestion relief, curtailment relief, and delivered energy value—exceed what ratepayers pay through transmission charges and energy losses. The CAISO Transmission Economic Assessment Methodology defines ratepayer benefit as the “difference of net load payment between the cases pre and post project” (CAISO 2017). The Brattle Group draws the same distinction: “from a ratepayer perspective, we calculate benefits and costs based on the additional transmission revenue requirements added to ratepayers’ bills, and the impacts on generation charges” (Pfeifenberger et al. 2021). Unlike the societal BCRs, the ratepayer numerator excludes avoided emissions ($B_{\text{avoided,emissions}}^P$) because they are a societal externality reduction that does not reduce wholesale energy purchase costs or appear on customer bills absent an embedded carbon price (Pfeifenberger et al. 2021), consistent with the ratepayer-impact-measure test principle (CPUC 2001). $BCR_{\text{rp}} < 1$ means ratepayers pay more (through rates and losses) than they receive in bill-relevant benefits; this is expected for projects whose primary value is emissions reduction or reliability.

Capital-only and Capital+Delay (BCR_{cap} , $BCR_{\text{cap+del}}$). The Capital-only BCR compares total benefits to capital investment alone, ignoring all lifecycle costs. Narrow benefit-cost analyses that use only capital costs in the denominator systematically overstate the investment case by missing operational, delay, risk, and emissions costs (Pfeifenberger et al. 2021). The CTCC includes this metric to demonstrate the magnitude of this effect. The Capital+Delay variant adds all delay costs (base, congestion, curtailment, and displacement delay), quantifying the total pre-COD plus capital exposure. FERC Order No. 1920 cites “lengthy and uncertain permitting and siting processes” as a key driver of transmission underinvestment (FERC 2024); the gap between the two capital screening BCRs measures the cost of that delay.

Exclusion variants. Three exclusion variants of the societal BCR are reported. Each answers a specific sensitivity question:

1. **Excl. avoided emissions:** FERC Order No. 1920 does not require emissions quantification as a transmission benefit; the seven required benefit categories reflect “objective, measurable changes in transmission system conditions, rather than achievement of state public policies” (FERC 2024). This variant shows the BCR under the FERC-minimum benefit set. It is the relevant metric for jurisdictions that do not recognize a social cost of carbon.
2. **Excl. wildfire:** Wildfire risk is geographically concentrated. Projects in low-wildfire regions (e.g., the Great Plains, Midwest) should not bear wildfire costs in their BCR.

3. **Excl. wildfire + outage:** Removes all probabilistic risk costs, showing the deterministic BCR for stakeholders who question probabilistic risk valuation methodology.

Additional exclusion combinations can be computed interactively using the CTCC’s custom BCR builder, which allows toggling any combination of five excludable cost/benefit groups (avoided emissions, loss-compensation emissions, line losses, wildfire risk, outage risk).

E Revenue

This section defines capital recovery—the depreciation and return-on-rate-base component of the full FERC revenue requirement. Capital recovery is a transfer such that the utility recovers its capital investment from ratepayers (it is not a cost or benefit to society). The full revenue requirement that enters the utility BCR also includes O&M and insurance (see the preceding paragraph on the full revenue requirement). Both perspectives use the same capital spending (same real-dollar amounts and timing). Utility (regulatory): base-year spending in CWIP is capitalized with AFUDC at real WACC to COD, giving RB_{real} . Society: same spending stream is discounted to base year at real WACC; AFUDC does not appear. Same inputs and time value; different purpose (rate base vs. PV of spending).

Variables

Variable	Description
RB_{real}	Rate base at COD, base-year (\$). AFUDC capital at COD (see §A.3). Already in base-year dollars.
C^{cap}	Total capital in rate base at COD (\$). $RB_{\text{real}} = C^{\text{cap}}$.
n	Project lifetime (years). Depreciation and revenue-requirement horizon; same as T_{lifetime} .
$r_{\text{WACC,real}}$	Real WACC (decimal). Derived from nominal WACC and inflation via the Fisher equation: $r_{\text{WACC,real}} = (1 + r_{\text{WACC,nominal}})/(1 + \pi) - 1$. Used for both the return component and discounting.
R_t	Annual revenue requirement in year t , real terms (\$/year). Straight-line depreciation plus return on the declining undepreciated balance; see equations below.
T_{lifetime}	Project lifetime (years).
$R_{\text{total,real}}$	Total revenue over lifetime, undiscounted, real (\$). Sum of R_t over $t = 1, \dots, n$; no time value.
R_{PV}	Present value of capital recovery (\$). Capital recovery component of the utility’s revenue requirement; combined with O&M, insurance, and ROW rent PV to form ATRR_{PV} .
T_{COD}	Commercial operation date (year index). Years from base year to first year of operation.

Equations

1. **Rate base.** RB_{real} is the AFUDC-capitalized rate base from §A.3, already in base-year dollars.

2. **COD (year index from base year).**

$$T_{\text{COD}} = T_{\text{delay}} + T_{\text{construction}} + 1.$$

3. **Annual revenue requirement.** Revenue follows the standard regulatory formulation: straight-line depreciation plus return on the undepreciated rate base at the real weighted-average cost of capital (FERC 2026; FERC 1995):

$$R_t = \frac{RB_{\text{real}}}{n} + RB_{\text{real}} \times r_{\text{WACC,real}} \times \left(1 - \frac{t-1}{n}\right), \quad t = 1, \dots, n.$$

The first term is straight-line depreciation (constant); the second is return on the declining undepreciated balance. Revenue declines from $R_1 = RB_{\text{real}}(1/n + r_{\text{WACC,real}})$ in year 1 to $R_n = RB_{\text{real}}/n$ in year n (Pfeifenberger et al. 2021).

4. **Total revenue over lifetime (real, undiscounted).**

$$R_{\text{total,real}} = \sum_{t=1}^n R_t.$$

5. **Present value of revenue (benefit to utility / cost to ratepayers).**

$$R_{\text{PV}} = \sum_{t=1}^n \frac{R_t}{(1 + r_{\text{WACC,real}})^{T_{\text{COD}} + t - 1}}.$$

By construction, $R_{\text{PV}} = RB_{\text{real}}/(1 + r_{\text{WACC,real}})^{T_{\text{COD}} - 1}$ — the NPV neutrality identity ensures the utility exactly recovers its investment plus a fair return (FERC 1995). So $B_{\text{utility}} = R_{\text{PV}}$ and $C_{\text{ratepayers}} = B_{\text{utility}} = R_{\text{PV}}$.

Full revenue requirement (utility perspective). The regulated utility’s annual revenue requirement follows FERC cost-of-service principles (FERC 2026):

$$\text{ATRR} = \underbrace{RB \times r + \text{Depreciation}}_{\text{capital recovery}} + \underbrace{C_{\text{O\&M}} + C_{\text{insurance}} + C_{\text{rent,total}}}_{\text{operating expenses}}.$$

O&M expenses (FERC Accounts 560–573) and property insurance (Account 924, allocated via Kansas-Nebraska gross-plant ratios) are recoverable operating expenses passed through to ratepayers at cost (*Cost-of-Service Rates Manual*, pp. 6, 19, 24). The utility BCR numerator includes all three components—capital recovery, O&M, and insurance—reflecting that a regulated utility recovers all prudently incurred costs through its formula rate. For a zero-delay project, $\text{BCR}_{\text{utility}} \approx 1.0$ by construction; deviation below 1.0 quantifies the present value of costs incurred during the pre-construction delay period that cannot be recovered through rates, since the asset is not yet “used and useful” per FERC’s rate-base inclusion standard.

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